## NAME

ovn-nb − OVN_Northbound database schema

This database is the interface between OVN and the cloud management system (CMS), such as OpenStack, running above it. The CMS produces almost all of the contents of the database. The **ovn−northd** program monitors the database contents, transforms it, and stores it into the **OVN_Southbound** database.

We generally speak of ''the'' CMS, but one can imagine scenarios in which multiple CMSes manage different parts of an OVN deployment.

### External IDs

Each of the tables in this database contains a special column, named **external_ids**. This column has the same form and purpose each place it appears.

**external_ids**: map of string-string pairs
Key-value pairs for use by the CMS. The CMS might use certain pairs, for example, to identify entities in its own configuration that correspond to those in this database.

## TABLE SUMMARY

The following list summarizes the purpose of each of the tables in the **OVN_Northbound** database. Each table is described in more detail on a later page.

| Table | Purpose |
|---|---|
| **NB_Global** | Northbound configuration |
| **Sample_Collector** | Sample_Collector |
| **Sample** | Sample |
| **Copp** | Control plane protection |
| **Logical_Switch** | L2 logical switch |
| **Logical_Switch_Port** | L2 logical switch port |
| **Logical_Switch_Port_Health_Check** | logical switch port health check |
| **Forwarding_Group** | forwarding group |
| **Address_Set** | Address Sets |
| **Port_Group** | Port Groups |
| **Load_Balancer** | load balancer |
| **Load_Balancer_Group** | load balancer group |
| **Load_Balancer_Health_Check** | load balancer |
| **ACL** | Access Control List (ACL) rule |
| **Logical_Router** | L3 logical router |
| **QoS** | QoS rule |
| **Mirror** | Mirror Entry |
| **Mirror_Rule** | Mirror rule entry |
| **Meter** | Meter entry |
| **Meter_Band** | Band for meter entries |
| **Logical_Router_Port** | L3 logical router port |
| **Logical_Router_Static_Route** | Logical router static routes |

**Logical_Router_Policy**
: Logical router policies

**NAT**
: NAT rules

**DHCP_Options**
: DHCP options

**DHCP_Relay**
: DHCP Relay

**Connection**
: OVSDB client connections.

**DNS**
: Native DNS resolution

**SSL**
: SSL configuration.

**Gateway_Chassis**
: Gateway_Chassis configuration.

**HA_Chassis_Group**
: HA_Chassis_Group configuration.

**HA_Chassis**
: HA_Chassis configuration.

**BFD**
: BFD configuration.

**Static_MAC_Binding**
: Static_MAC_Binding configuration.

**Chassis_Template_Var**
: Chassis_Template_Var configuration.

**Sampling_App**
: Sampling_App configuration.

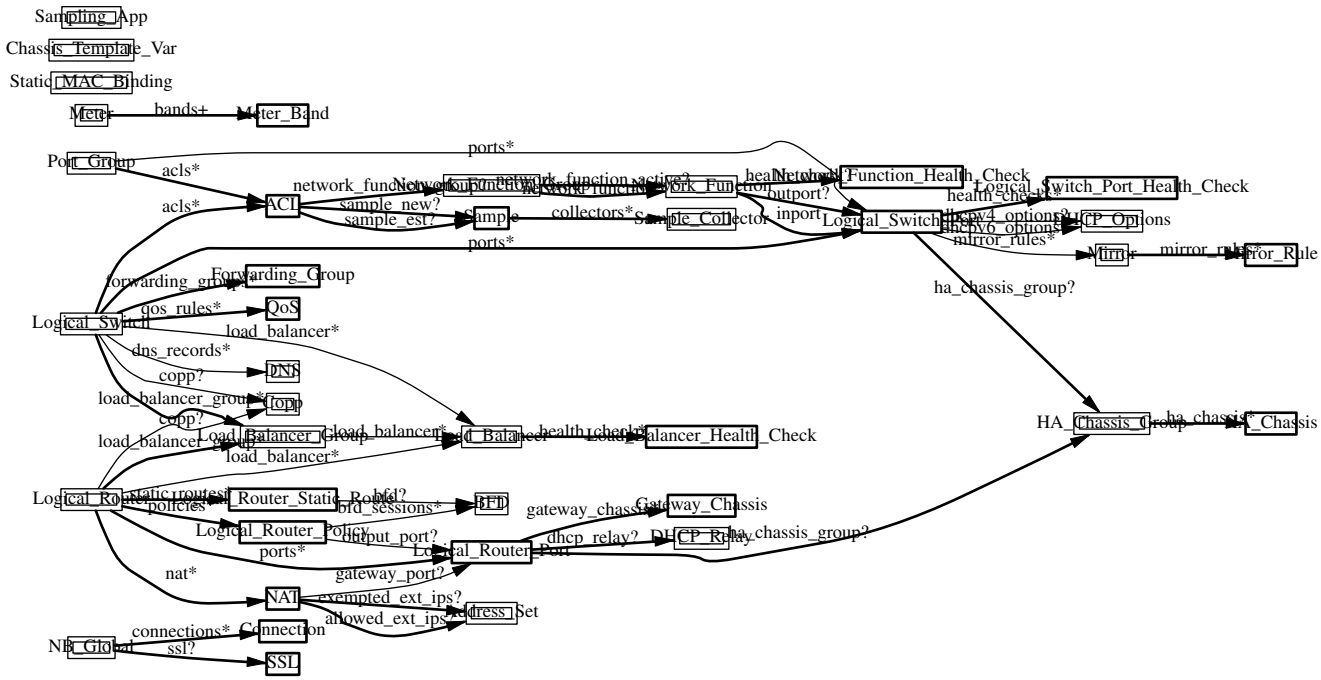
**Network_Function_Group**
: network function group

**Network_Function**
: network function

**Network_Function_Health_Check**
: network function health check

## TABLE RELATIONSHIPS

The following diagram shows the relationship among tables in the database. Each node represents a table. Tables that are part of the ‘‘root set’’ are shown with double borders. Each edge leads from the table that contains it and points to the table that its value represents. Edges are labeled with their column names, followed by a constraint on the number of allowed values: **?** for zero or one, ***** for zero or more, **+** for one or more. Thick lines represent strong references; thin lines represent weak references.



## NB_Global TABLE

Northbound configuration for an OVN system. This table must have exactly one row.

**Summary:**

| | |
|---|---|
| *Identity:* | |
| **name** | string |
| *Status:* | |
| **nb_cfg** | integer |
| **nb_cfg_timestamp** | integer |
| **sb_cfg** | integer |
| **sb_cfg_timestamp** | integer |
| **hv_cfg** | integer |
| **hv_cfg_timestamp** | integer |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |
| *Common options:* | |
| **options** | map of string-string pairs |
| *Options for configuring OVS BFD:* | |
| **options : bfd-min-rx** | optional string |
| **options : bfd-decay-min-rx** | optional string |
| **options : bfd-min-tx** | optional string |
| **options : bfd-mult** | optional string |
| **options : ignore_chassis_features** | optional string |
| **options : mac_prefix** | optional string |
| **options : mac_binding_removal_limit** | optional string, containing an integer, in range 0 to 4,294,967,295 |
| **options : fdb_removal_limit** | optional string, containing an integer, in range 0 to 4,294,967,295 |
| **options : controller_event** | optional string, either **true** or **false** |
| **options : northd_probe_interval** | optional string |
| **options : ic_probe_interval** | optional string |
| **options : nbctl_probe_interval** | optional string |
| **options : northd_trim_timeout** | optional string |
| **options : use_logical_dp_groups** | optional string |
| **options : use_parallel_build** | optional string |
| **options : ignore_lsp_down** | optional string |
| **options : use_ct_inv_match** | optional string |
| **options : acl_ct_translation** | optional string |
| **options : default_acl_drop** | optional string |
| **options : debug_drop_domain_id** | optional string |
| **options : debug_drop_collector_set** | optional string |
| **options : use_common_zone** | optional string, either **true** or **false** |
| **options : northd-backoff-interval-ms** | optional string |
| **options : vxlan_mode** | optional string |
| **options : always_tunnel** | optional string, either **true** or **false** |
| **options : ecmp_nexthop_monitor_enable** | optional string |
| **options : enable_chassis_nb_cfg_update** | optional string |
| *Options for services health check confguration:* | |
| **options : svc_monitor_mac** | optional string |
| **options : svc_monitor_mac_dst** | optional string |
| **options : svc_monitor_ip** | optional string |
| **options : svc_monitor_ip_dst** | optional string |

*Options for configuring interconnection route advertisement:*

| | |
|---|---|
| **options : ic-route-adv** | optional string |
| **options : ic-route-learn** | optional string |
| **options : ic-route-adv-default** | optional string |
| **options : ic-route-learn-default** | optional string |
| **options : ic-route-adv-lb** | optional string |
| **options : ic-route-learn-lb** | optional string |
| **options : ic-route-denylist** | optional string |

*Options for interconnection address_set advertisement:*

| | |
|---|---|
| **options : ic-as-adv** | optional string |
| **options : ic-as-learn** | optional string |

*Connection Options:*

| | |
|---|---|
| **connections** | set of **Connection**s |
| **ssl** | optional **SSL** |

*Security Configurations:*

| | |
|---|---|
| **ipsec** | boolean |
| **options : ipsec_*** | optional string |

*Read-only Options:*

| | |
|---|---|
| **options : max_tunid** | optional string |

**Details:**

*Identity:*

**name**: string
: The name of the OVN cluster, which uniquely identifies the OVN cluster throughout all OVN clusters supposed to interconnect with each other.

*Status:*

These columns allow a client to track the overall configuration state of the system.

**nb_cfg**: integer
: Sequence number for client to increment. When a client modifies any part of the northbound database configuration and wishes to wait for **ovn−northd** and possibly all of the hypervisors to finish applying the changes, it may increment this sequence number.

**nb_cfg_timestamp**: integer
: The timestamp, in milliseconds since the epoch, when **ovn−northd** sees the latest **nb_cfg** and starts processing.

  To print the timestamp as a human-readable date:

  **date −d "@$(ovn−nbctl get NB_Global . nb_cfg_timestamp | sed 's/...$//')"**

**sb_cfg**: integer
: Sequence number that **ovn−northd** sets to the value of **nb_cfg** after it finishes applying the corresponding configuration changes to the **OVN_Southbound** database.

**sb_cfg_timestamp**: integer
: The timestamp, in milliseconds since the epoch, when **ovn−northd** finishes applying the corresponding configuration changes to the **OVN_Southbound** database successfully.

**hv_cfg**: integer
: Sequence number that **ovn−northd** sets to the smallest sequence number of all the chassis in the system, as reported in the **Chassis_Private** table in the southbound database. Thus, **hv_cfg** equals **nb_cfg** if all chassis are caught up with the northbound configuration (which may never happen, if any chassis is down). This value can regress, if a chassis was removed from the system and rejoins before catching up.

If there are no chassis, then **ovn−northd** copies **nb_cfg** to **hv_cfg**. Thus, in this case, the (nonexistent) hypervisors are always considered to be caught up. This means that hypervisors can be "caught up" even in cases where **sb_cfg** would show that the southbound database is not. To detect when both the hypervisors and the southbound database are caught up, a client should take the smaller of **sb_cfg** and **hv_cfg**.

**hv_cfg_timestamp**: integer
The largest timestamp, in milliseconds since the epoch, of the smallest sequence number of all the chassis in the system, as reported in the **Chassis_Private** table in the southbound database. In other words, this timestamp reflects the time when the slowest chassis catches up with the northbound configuration, which is useful for end-to-end control plane latency measurement.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

*Common options:*

**options**: map of string-string pairs
This column provides general key/value settings. The supported options are described individually below.

*Options for configuring OVS BFD:*

These options apply when **ovn−controller** configures OVS BFD on tunnels interfaces. Please note these parameters refer to legacy OVS BFD implementation and not to OVN BFD one.

**options : bfd-min-rx**: optional string
BFD option **min−rx** value to use when configuring BFD on tunnel interfaces.

**options : bfd-decay-min-rx**: optional string
BFD option **decay−min−rx** value to use when configuring BFD on tunnel interfaces.

**options : bfd-min-tx**: optional string
BFD option **min−tx** value to use when configuring BFD on tunnel interfaces.

**options : bfd-mult**: optional string
BFD option **mult** value to use when configuring BFD on tunnel interfaces.

**options : ignore_chassis_features**: optional string
When set to **false**, the **ovn−northd** will evaluate the features supported by each chassis and will only activate features that are universally supported by all chassis. This approach is crucial for maintaining backward compatibility during an upgrade when the **ovn−northd** is updated prior to the **ovn−controller**. However, if any chassis is poorly managed and the upgrade is unsuccessful, it will restrict **ovn−northd** from activating the new features.

Alternatively, setting this option to **true** instructs **ovn−northd** to bypass the support status of features on each chassis and to directly implement the latest features. This approach safeguards the operation of **ovn−northd** from being adversely affected by a mismatched configuration of a chassis.

The default setting for this option is **false**.

**options : mac_prefix**: optional string
Configure a given OUI to be used as prefix when L2 address is dynamically assigned, e.g. **00:11:22**

**options : mac_binding_removal_limit**: optional string, containing an integer, in range 0 to 4,294,967,295
MAC binding aging bulk removal limit. This limits how many rows can expire in a single transaction. Default value is 0 which is unlimited. When we hit the limit next batch removal is delayed by 5 s.

**options : fdb_removal_limit**: optional string, containing an integer, in range 0 to 4,294,967,295
FDB aging bulk removal limit. This limits how many rows can expire in a single transaction. Default value is 0 which is unlimited. When we hit the limit next batch removal is delayed by 5 s.

**options : controller_event**: optional string, either **true** or **false**
Value set by the CMS to enable/disable ovn-controller event reporting. Traffic into OVS can raise a 'controller' event that results in a Controller_Event being written to the **Controller_Event** table in SBDB. When the CMS has seen the event and taken appropriate action, it can remove the corresponding row in **Controller_Event** table. The intention is for a CMS to see the events and take some sort of action. Please see the **Controller_Event** table in SBDB. It is possible to associate a meter to each controller event type in order to not overload the pinctrl thread under heavy load. Each event type relies on a meter with a defined name:

- empty_lb_backends: event-elb

**options : northd_probe_interval**: optional string
The inactivity probe interval of the connection to the OVN Northbound and Southbound databases from **ovn−northd**, in milliseconds. If the value is zero, it disables the connection keepalive feature.

If the value is nonzero, then it will be forced to a value of at least 1000 ms.

**options : ic_probe_interval**: optional string
The inactivity probe interval of the connection to the OVN Northbound and Southbound databases from **ovn−ic**, in milliseconds. If the value is zero, it disables the connection keepalive feature.

If the value is nonzero, then it will be forced to a value of at least 1000 ms.

**options : nbctl_probe_interval**: optional string
The inactivity probe interval of the connection to the OVN Northbound database from **ovn−nbctl** utility, in milliseconds. If the value is zero, it disables the connection keepalive feature.

If the value is nonzero, then it will be forced to a value of at least 1000 ms.

If the value is less than zero, then the default inactivity probe interval for **ovn−nbctl** would be left intact (120000 ms).

**options : northd_trim_timeout**: optional string
When used, this configuration value specifies the time, in milliseconds, since the last **ovn−northd** active operation after which memory trimming is performed. By default this is set to 30000 (30 seconds).

**options : use_logical_dp_groups**: optional string
Note: This option is deprecated, the only behavior is to always combine logical flows by datapath groups. Changing the value or removing this option all toghether will have no effect.

**ovn−northd** combines logical flows that differs only by logical datapath into a single logical flow with logical datapath group attached.

**options : use_parallel_build**: optional string
If set to **true**, **ovn−northd** will attempt to compute logical flows in parallel.

Parallel computation is enabled only if the system has 4 or more cores/threads available to be used by ovn-northd.

The default value is **false**.

**options : ignore_lsp_down**: optional string
If set to false, ARP/ND reply flows for logical switch ports will be installed only if the port is up, i.e. claimed by a Chassis. If set to true, these flows are installed regardless of the status of the port, which can result in a situation that ARP request to an IP is resolved even before the relevant VM/container is running. For environments where this is not an issue, setting it to **true** can reduce the load and latency of the control plane. The default value is **true**.

**options : use_ct_inv_match**: optional string
If set to false, **ovn−northd** will not use the **ct.inv** field in any of the logical flow matches. The default value is true. If the NIC supports offloading OVS datapath flows but doesn't support offloading ct_state **inv** flag, then the datapath flows matching on this flag (either **+inv** or **−inv**) will not be

offloaded. CMS should consider setting **use_ct_inv_match** to **false** in such cases. This results in a side effect of the invalid packets getting delivered to the destination VIF, which otherwise would have been dropped by **OVN**.

**options : acl_ct_translation**: optional string
If set to **true**, **ovn−northd** will enable connection tracking based L4 field translation for stateful ACLs. When enabled, ACL matches on L4 port fields (tcp/udp/sctp) use connection tracking state instead of packet headers. This allows proper handling of IP fragmentation in userspace datapaths (e.g., DPDK) where fragments are not automatically reassembled during connection tracking lookup.

**Important:** Enabling this option may break hardware offloading of flows. It also disables the multicast conntrack bypass, which means multicast traffic (including DHCP) will go through connection tracking. This may have a performance impact on multicast-heavy workloads. Only enable this option if you need to handle fragmented traffic with stateful ACLs in userspace datapaths.

The default value is **false**.

**options : default_acl_drop**: optional string
If set to **true**., **ovn−northd** will generate a logical flow to drop all traffic in the ACL stages. By default this option is set to **false**.

**options : debug_drop_domain_id**: optional string
If set to a 8-bit number and if **debug_drop_collector_set** is also configured, **ovn−northd** will add a **sample** action to every logical flow that contains a 'drop' action. The 8 most significant bits of the observation_domain_id field will be those specified in the **debug_drop_domain_id**. The 24 least significant bits of the observation_domain_id field will be the datapath's key.

The observation_point_id will be set to the first 32 bits of the logical flow's UUID.

Note: This key is deprecated in favor of the value configured in the **Sampling_App** table for the **drop** application.

**options : debug_drop_collector_set**: optional string
If set to a 32-bit number **ovn−northd** will add a **sample** action to every logical flow that contains a 'drop' action. The sample action will have the specified collector_set_id. The value must match that of the local OVS configuration as described in **ovs−actions**(7).

**options : use_common_zone**: optional string, either **true** or **false**
Default value is **false**. If set to **true** the SNAT and DNAT happens in common zone, instead of happening in separate zones, depending on the configuration. However, this option breaks traffic when there is configuration of DGP + LB + SNAT on this LR. The value **true** should be used only in case of HWOL compatibility with GDP.

**options : northd-backoff-interval-ms**: optional string
Maximum interval that the northd incremental engine is delayed by in milliseconds. Setting the value to nonzero delays the next northd engine run by the previous run time, capped by the specified value. If the value is zero the engine won't be delayed at all. The recommended period is smaller than 500 ms, beyond that the latency of SB changes would be very noticeable.

**options : vxlan_mode**: optional string
By default if at least one chassis in OVN cluster has VXLAN encap, northd will run in a **VXLAN mode**. See man ovn-architecture(7) **Tunnel Encapsulations** paragraph for more details. In case VXLAN encaps are needed on chassis only to support HW VTEP functionality and main encap type is GENEVE, set this option to **false** to use default non-**VXLAN mode** tunnel IDs allocation logic. Please consider when OVN is running in **OVN−interconnect** mode and it is using **VXLAN** encapsulation type, the max number of non-transit logical switches and logical routers is reduced to 1024. Plese note, in order to enable **VXLAN** encapsulation type for for cross-AZ traffic, **vxlan_mode** parameter in **IC_NB_Global TABLE** must be set to true.

**options : always_tunnel**: optional string, either **true** or **false**
If set to true, then the traffic destined to a VIF of a provider logical switch (having a localnet port) will be tunnelled instead of sending it via the localnet port. This option will be useful if CMS wants to connect overlay logical switches (without localnet port) and provider logical switches to a router. Without this option set, the traffic path will be a mix of tunnelling and localnet ports (since routing is distributed) resulting in the leakage of the router port mac address to the upstream switches and undefined behavior if NATting is involed. This option is disabled by default.

**options : ecmp_nexthop_monitor_enable**: optional string
If set to **true**., **ovn−northd** will create entries in **ECMP_Nexthop TABLE** to track ECMP routes created with **−−ecmp_symmetric_reply** option. By default this option is set to **false**.

**options : enable_chassis_nb_cfg_update**: optional string
If set to **false**, ovn-controllers will no longer update the **nb_cfg** column in the **Chassis_Private** table of the **OVN_Southbound** database. They will still update the **external_ids:ovn−nb−cfg** in the local OVS integration bridge. By defailt this option is set to **true**.

*Options for services health check confguration:*

These options are used when health configuration is enabled for **Load_Balancer** and **Network_Function** services.

**options : svc_monitor_mac**: optional string
MAC Address used as the Ethernet source in health check probes. If unspecified, a MAC address is automatically generated.

**options : svc_monitor_mac_dst**: optional string
MAC Address used as the Ethernet destination in health check probes. If unspecified, a MAC address is automatically generated. This applies only to Network Function health check probes deployed in inline mode.

**options : svc_monitor_ip**: optional string
IP Address (IPv4 or IPv6) used as the source in health check probes. This applies only to Network Function health check probes deployed in inline mode.

**options : svc_monitor_ip_dst**: optional string
IP Address (IPv4 or IPv6) used as the destination in health check probes. This applies only to Network Function health check probes deployed in inline mode.

*Options for configuring interconnection route advertisement:*

These options control how routes are advertised between OVN deployments for interconnection. If enabled, **ovn−ic** from different OVN deployments exchanges routes between each other through the global **OVN_IC_Southbound** database. Only routers with ports connected to interconnection transit switches participate in route advertisement. For each of these routers, there are three types of routes to be advertised:

Firstly, the static routes configured in the router are advertised.

Secondly, the **networks** configured in the logical router ports that are not on the transit switches are advertised. These are considered as directly connected subnets on the router.

Thirdly, the vips of the load balancers associated with the logical router are advertised.

Link local prefixes (IPv4 169.254.0.0/16 and IPv6 FE80::/10) are never advertised.

The learned routes are added to the **static_routes** column of the **Logical_Router** table, with **external_ids:ic−learned−route** set to the uuid of the row in **Route** table of the **OVN_IC_Southbound** database.

**options : ic-route-adv**: optional string
A boolean value that enables route advertisement to the global **OVN_IC_Southbound** database. Default is **false**.

**options : ic-route-learn**: optional string
A boolean value that enables route learning from the global **OVN_IC_Southbound** database. Default is **false**.

**options : ic-route-adv-default**: optional string
A boolean value that enables advertising default route to the global **OVN_IC_Southbound** database. Default is **false**. This option takes effect only when option **ic−route−adv** is **true**.

**options : ic-route-learn-default**: optional string
A boolean value that enables learning default route from the global **OVN_IC_Southbound** database. Default is **false**. This option takes effect only when option **ic−route−learn** is **true**.

**options : ic-route-adv-lb**: optional string
A boolean value that enables advertising routes for load balancer vips to the global **OVN_IC_Southbound** database. Default is **false**. This option takes effect only when option **ic−route−adv** is **true**.

**options : ic-route-learn-lb**: optional string
A boolean value that enables learning routes for load balancer routes from the global **OVN_IC_Southbound** database. Default is **false**. This option takes effect only when option **ic−route−learn** is **true**.

**options : ic-route-denylist**: optional string
A string value contains a list of CIDRs delimited by ",". A route will not be advertised or learned if the route's prefix belongs to any of the CIDRs listed.

*Options for interconnection address_set advertisement:*

These options control whether address_sets propagate from one OVN deployment to another via the global **OVN_IC_Southbound**.

**options : ic-as-adv**: optional string
A boolean value that enables address_set advertisement to the global **OVN_IC_Southbound** database. Default is **false**.

**options : ic-as-learn**: optional string
A boolean value that enables address_set learning from the global **OVN_IC_Southbound** database. Default is **false**.

*Connection Options:*

**connections**: set of **Connection**s
Database clients to which the Open vSwitch database server should connect or on which it should listen, along with options for how these connections should be configured. See the **Connection** table for more information.

**ssl**: optional **SSL**
Global SSL/TLS configuration.

*Security Configurations:*

**ipsec**: boolean
Tunnel encryption configuration. If this column is set to be true, all OVN tunnels will be encrypted with IPsec.

**options : ipsec_***: optional string
IPsec configuration parameters are passed to IPsec backend by prefixing libreswan/strongswan options with ipsec_. Please reference **ipsec.conf**(5) for a comprehensive set of instructions on IPsec configuration.

*Read-only Options:*

**options : max_tunid**: optional string
The maximum supported tunnel ID. Depends on types of encapsulation enabled in the cluster.

### Sample_Collector TABLE

**Summary:**

| | |
|---|---|
| **id** | integer, in range 1 to 255 (must be unique within table) |
| **name** | string |
| **probability** | integer, in range 0 to 65,535 |
| **set_id** | integer, in range 1 to 4,294,967,295 |
| **external_ids** | map of string-string pairs |

**Details:**

**id**: integer, in range 1 to 255 (must be unique within table)

Sample collector unique id used for differentiating collectors that use the same **set_id** with different **probability** values. The supported value range for IDs is **1−255**.

**name**: string

Name of the sample collector.

**probability**: integer, in range 0 to 65,535

Sampling probability for this collector. It must be an integer number between 0 and 65535. A value of 0 corresponds to no packets being sampled while a value of 65535 corresponds to all packets being sampled.

**set_id**: integer, in range 1 to 4,294,967,295

The 32-bit integer identifier of the set of of collectors to send packets to. See Flow_Sample_Collector_Set Table in ovs-vswitchd's database schema.

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### Sample TABLE

This table describes a Sampling configuration. Entries in other tables might be associated with Sample entries to indicate how the sample should be generated. For an example, see **ACL**.

**Summary:**

| | |
|---|---|
| **collectors** | set of **Sample_Collector**s |
| **metadata** | integer, in range 1 to 4,294,967,295 (must be unique within table) |

**Details:**

**collectors**: set of **Sample_Collector**s

A list of references to **Sample_Collector** records to be used when generating samples (e.g., IPFIX). A sample can be sent to multiple collectors simultaneously.

**metadata**: integer, in range 1 to 4,294,967,295 (must be unique within table)

Will be used as Observation Point ID in every sample. The Observation Domain ID will be generated by ovn-northd and includes the logical datapath key as the least significant 24 bits and the sampling application type (e.g., drop debugging) as the 8 most significant bits.

### Copp TABLE

This table is used to define control plane protection policies, i.e., associate entries from table **Meter** to control protocol names.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **meters : arp** | optional string |
| **meters : arp-resolve** | optional string |
| **meters : dhcpv4-opts** | optional string |
| **meters : dhcpv6-opts** | optional string |
| **meters : dns** | optional string |
| **meters : event-elb** | optional string |
| **meters : icmp4-error** | optional string |
| **meters : icmp6-error** | optional string |
| **meters : igmp** | optional string |
| **meters : nd-na** | optional string |
| **meters : nd-ns** | optional string |
| **meters : nd-ns-resolve** | optional string |
| **meters : nd-ra-opts** | optional string |
| **meters : tcp-reset** | optional string |
| **meters : bfd** | optional string |
| **meters : reject** | optional string |
| **meters : svc-monitor** | optional string |
| **meters : dhcpv4-relay** | optional string |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)
CoPP name.

**meters : arp**: optional string
Rate limiting meter for ARP packets (request/reply) used for learning neighbors.

**meters : arp-resolve**: optional string
Rate limiting meter for packets that require resolving the next-hop (through ARP).

**meters : dhcpv4-opts**: optional string
Rate limiting meter for packets that require adding DHCPv4 options.

**meters : dhcpv6-opts**: optional string
Rate limiting meter for packets that require adding DHCPv6 options.

**meters : dns**: optional string
Rate limiting meter for DNS query packets that need to be replied to.

**meters : event-elb**: optional string
Rate limiting meter for empty load balancer events.

**meters : icmp4-error**: optional string
Rate limiting meter for packets that require replying with an ICMP error.

**meters : icmp6-error**: optional string
Rate limiting meter for packets that require replying with an ICMPv6 error.

**meters : igmp**: optional string
Rate limiting meter for IGMP packets.

**meters : nd-na**: optional string
Rate limiting meter for ND neighbor advertisement packets used for learning neighbors.

**meters : nd-ns**: optional string
Rate limiting meter for ND neighbor solicitation packets used for learning neighbors.

**meters : nd-ns-resolve**: optional string
Rate limiting meter for packets that require resolving the next-hop (through ND).

**meters : nd-ra-opts**: optional string
Rate limiting meter for packets that require adding ND router advertisement options.

**meters : tcp-reset**: optional string
Rate limiting meter for packets that require replying with TCP RST packet.

**meters : bfd**: optional string
Rate limiting meter for BFD packets.

**meters : reject**: optional string
Rate limiting meter for packets that trigger a reject action

**meters : svc-monitor**: optional string
Rate limiting meter for packets that are arriving to service monitor MAC address.

**meters : dhcpv4-relay**: optional string
Rate limiting meter for DHCPv4 relay packets (request/response) when DHCPv4 Relay functionality is enabled.

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

### Logical_Switch TABLE

Each row represents one L2 logical switch.

There are two kinds of logical switches, that is, ones that fully virtualize the network (overlay logical switches) and ones that provide simple connectivity to physical networks (bridged logical switches). They work in the same way when providing connectivity between logical ports on same chassis, but differently when connecting remote logical ports. Overlay logical switches connect remote logical ports by tunnels, while bridged logical switches provide connectivity to remote ports by bridging the packets to directly connected physical L2 segments with the help of **localnet** ports. Each bridged logical switch has one or more **localnet** ports, which have only one special address **unknown**.

**Summary:**

| | |
|---|---|
| **ports** | set of **Logical_Switch_Port**s |
| **load_balancer** | set of weak reference to **Load_Balancer**s |
| **load_balancer_group** | set of **Load_Balancer_Group**s |
| **acls** | set of **ACL**s |
| **qos_rules** | set of **QoS**es |
| **dns_records** | set of weak reference to **DNS**es |
| **forwarding_groups** | set of **Forwarding_Group**s |
| *Naming:* | |
| **name** | string |
| **external_ids : neutron:network_name** | optional string |
| *IP Address Assignment:* | |
| **other_config : subnet** | optional string |
| **other_config : exclude_ips** | optional string |
| **other_config : ipv6_prefix** | optional string |
| **other_config : dhcp_relay_port** | optional string |
| **other_config : mac_only** | optional string, either **true** or **false** |
| **other_config : fdb_age_threshold** | optional string, containing an integer, in range 0 to 4,294,967,295 |
| **other_config : ct-zone-limit** | optional string, containing an integer, in range 0 to 4,294,967,295 |
| **other_config : enable-stateless-acl-with-lb** | optional string, either **true** or **false** |
| *EVPN Options:* | |
| **other_config : dynamic-routing-vni** | optional string, containing an integer, in range 0 to 16,777,215 |
| **other_config : dynamic-routing-bridge-ifname** | optional string |
| **other_config : dynamic-routing-vxlan-ifname** | optional string |
| **other_config : dynamic-routing-advertise-ifname** | optional string |
| **other_config : dynamic-routing-fdb-prefer-local** | optional string, either **true** or **false** |
| **other_config : dynamic-routing-redistribute** | optional string |
| *IP Multicast Snooping Options:* | |
| **other_config : mcast_snoop** | optional string, either **true** or **false** |
| **other_config : mcast_querier** | optional string, either **true** or **false** |
| **other_config : mcast_flood_unregistered** | optional string, either **true** or **false** |
| **other_config : mcast_table_size** | optional string, containing an integer, in range 1 to 32,766 |
| **other_config : mcast_idle_timeout** | optional string, containing an integer, in range 15 to 3,600 |

| | |
|---|---|
| **other_config : mcast_query_interval** | optional string, containing an integer, in range 1 to 3,600 |
| **other_config : mcast_query_max_response** | optional string, containing an integer, in range 1 to 10 |
| **other_config : mcast_eth_src** | optional string |
| **other_config : mcast_ip4_src** | optional string |
| **other_config : mcast_ip6_src** | optional string |
| *Interconnection:* | |
| **other_config : interconn-ts** | optional string |
| **other_config : ic-vxlan_mode** | optional string, either **true** or **false** |
| *Tunnel Key:* | |
| **other_config : requested-tnl-key** | optional string, containing an integer, in range 1 to 16,777,215 |
| **copp** | optional weak reference to **Copp** |
| *Other options:* | |
| **other_config : vlan-passthru** | optional string, either **true** or **false** |
| **other_config : broadcast-arps-to-all-routers** | optional string, either **true** or **false** |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**ports**: set of **Logical_Switch_Port**s

The logical ports connected to the logical switch.

It is an error for multiple logical switches to include the same logical port.

**load_balancer**: set of weak reference to **Load_Balancer**s

Set of load balancers associated to this logical switch.

**load_balancer_group**: set of **Load_Balancer_Group**s

Set of load balancers groups associated to this logical switch.

**acls**: set of **ACL**s

Access control rules that apply to packets within the logical switch.

**qos_rules**: set of **QoS**es

QoS marking and metering rules that apply to packets within the logical switch.

**dns_records**: set of weak reference to **DNS**es

This column defines the DNS records to be used for resolving internal DNS queries within the logical switch by the native DNS resolver. Please see the **DNS** table.

**forwarding_groups**: set of **Forwarding_Group**s

Groups a set of logical port endpoints for traffic going out of the logical switch.

*Naming:*

These columns provide names for the logical switch. From OVN's perspective, these names have no special meaning or purpose other than to provide convenience for human interaction with the database. There is no requirement for the name to be unique. (For a unique identifier for a logical switch, use its row UUID.)

(Originally, **name** was intended to serve the purpose of a human-friendly name, but the Neutron integration used it to uniquely identify its own switch object, in the format **neutron–***uuid*. Later on, Neutron started propagating the friendly name of a switch as **external_ids:neutron:network_name**. Perhaps this can be cleaned up someday.)

**name**: string

A name for the logical switch.

**external_ids : neutron:network_name**: optional string

Another name for the logical switch.

*IP Address Assignment:*

These options control automatic IP address management (IPAM) for ports attached to the logical switch. To enable IPAM for IPv4, set **other_config:subnet** and optionally **other_config:exclude_ips**. To enable IPAM for IPv6, set **other_config:ipv6_prefix**. IPv4 and IPv6 may be enabled together or separately.

To request dynamic address assignment for a particular port, use the **dynamic** keyword in the **addresses** column of the port's **Logical_Switch_Port** row. This requests both an IPv4 and an IPv6 address, if IPAM for IPv4 and IPv6 are both enabled.

**other_config : subnet**: optional string
: Set this to an IPv4 subnet, e.g. **192.168.0.0/24**, to enable **ovn−northd** to automatically assign IP addresses within that subnet.

**other_config : exclude_ips**: optional string
: To exclude some addresses from automatic IP address management, set this to a list of the IPv4 addresses or **..**-delimited ranges to exclude. The addresses or ranges should be a subset of those in **other_config:subnet**.

  Whether listed or not, **ovn−northd** will never allocate the first or last address in a subnet, such as 192.168.0.0 or 192.168.0.255 in 192.168.0.0/24.

  Examples:

  - **192.168.0.2 192.168.0.10**
  - **192.168.0.4 192.168.0.30..192.168.0.60 192.168.0.110..192.168.0.120**
  - **192.168.0.110..192.168.0.120 192.168.0.25..192.168.0.30 192.168.0.144**

**other_config : ipv6_prefix**: optional string
: Set this to an IPv6 prefix to enable **ovn−northd** to automatically assign IPv6 addresses using this prefix. The assigned IPv6 address will be generated using the IPv6 prefix and the MAC address (converted to an IEEE EUI64 identifier) of the port. The IPv6 prefix defined here should be a valid IPv6 address ending with **::**.

  Examples:

  - **aef0::**
  - **bef0:1234:a890:5678::**
  - **8230:5678::**

**other_config : dhcp_relay_port**: optional string
: If set to the name of logical switch port of type **router** then, DHCP Relay is enabled for this logical switch provided the corresponding **Logical_Router_Port** has DHCP Relay configured.

**other_config : mac_only**: optional string, either **true** or **false**
: Value used to request to assign L2 address only if neither subnet nor ipv6_prefix are specified

**other_config : fdb_age_threshold**: optional string, containing an integer, in range 0 to 4,294,967,295
: FDB aging **threshold** value in seconds. FDB exceeding this timeout will be automatically removed. The value defaults to 0, which means disabled.

**other_config : ct-zone-limit**: optional string, containing an integer, in range 0 to 4,294,967,295
: CT zone **limit** value for given **Logical_Switch**. This value will be propagated to all **Logical_Switch_Port** when configured, but can be overwritten individually per **Logical_Switch_Port**. The value 0 means unlimited. When the option is not present the limit is not set and the zone limit is derived from OvS default datapath limit.

**other_config : enable-stateless-acl-with-lb**: optional string, either **true** or **false**
: This option must be set to true for stateless ACL to work with load balancers. When enabled, packets with the ct.inv flag will not be dropped, even if use_ct_inv_match is set to true. Default: **false**.

*EVPN Options:*

These options control the EVPN configuration of the logical switch. To enable EVPN integration set **other_config:dynamic-routing-vni** to a valid VNI number to represent the EVPN L2 domain extended by the logical switch.

**other_config : dynamic-routing-vni**: optional string, containing an integer, in range 0 to 16,777,215
This defines the vni number associated with EVPN domain that the Logical Switch is supposed to connect to.

All of the following configuration options must also be provided in order for the configuration to be valid: **other_config:dynamic-routing-bridge-ifname**, **other_config:dynamic-routing-vxlan-ifname**, **other_config:dynamic-routing-advertise-ifname**.

**other_config : dynamic-routing-bridge-ifname**: optional string
Set the interface name for the bridge used for EVPN integration. Only relevant if **other_config:dynamic-routing-vni** is set to valid VNI.

**other_config : dynamic-routing-vxlan-ifname**: optional string
List of interface names used for the vxlan devices used for EVPN integration. Only relevant if **other_config:dynamic-routing-vni** is set to valid VNI.

**other_config : dynamic-routing-advertise-ifname**: optional string
Set the interface name for the advertise device used for EVPN integration. Only relevant if **other_config:dynamic-routing-vni** is set to valid VNI.

**other_config : dynamic-routing-fdb-prefer-local**: optional string, either **true** or **false**
This option defines the preference of FDB lookup, if set to true OVN will try to find the FDB entry in SB **FDB** table first. Then it tries to resolve the FDB via **ovn−controller** local EVPN FDB cache. The option defaults to false.

Only relevant if **other_config:dynamic-routing-vni** is set to valid VNI.

**other_config : dynamic-routing-redistribute**: optional string
Only relevant if **other_config:dynamic-routing-vni** is set to valid VNI.

If **fdb** is specified then ovn-controller will advertise all workloads that are local to the chassis. This applies to VIFs, container ports, virtual ports, connected DGPs and connected GW routers. Additionally, dynamically learned FDB entries (from ports with **unknown** addresses) are also advertised. For VIF ports, only entries learned on the local chassis are advertised; for localnet ports, entries are advertised on all chassis where the datapath is local.

If **ip** is specified then ovn-controller will advertise all IPs/MACs bindings that are local to the chassis. This applies to VIFs and router ports.

If **nat** is specified then the external IP/MAC of any distributed **dnat_and_snat** NAT entry (for example a floating IP) whose distributed gateway port is attached to this logical switch is advertised as an EVPN Type−2 route, using the NAT's **external_mac**. This is independent of **ip**: enabling **ip** alone does not advertise floating IPs, and enabling **nat** alone does not advertise VIF or router-port addresses. This mirrors the **nat** token of **options:dynamic-routing-redistribute**, which advertises the same NAT entries as EVPN Type−5 routes.

*IP Multicast Snooping Options:*

These options control IP Multicast Snooping configuration of the logical switch. To enable IP Multicast Snooping set **other_config:mcast_snoop** to true. To enable IP Multicast Querier set **other_config:mcast_querier** to true. If IP Multicast Querier is enabled **other_config:mcast_eth_src** and **other_config:mcast_ip4_src** must be set.

**other_config : mcast_snoop**: optional string, either **true** or **false**
Enables/disables IP Multicast Snooping on the logical switch. Default: **false**.

**other_config : mcast_querier**: optional string, either **true** or **false**
Enables/disables IP Multicast Querier on the logical switch. Only applicable if **other_config:mcast_snoop** is enabled. Default: **true**.

**other_config : mcast_flood_unregistered**: optional string, either **true** or **false**
Determines whether unregistered multicast traffic should be flooded or not. Only applicable if **other_config:mcast_snoop** is enabled. Default: **false**.

**other_config : mcast_table_size**: optional string, containing an integer, in range 1 to 32,766
Number of multicast groups to be stored. Default: 2048.

**other_config : mcast_idle_timeout**: optional string, containing an integer, in range 15 to 3,600
Configures the IP Multicast Snooping group idle timeout (in seconds). Default: 300 seconds.

**other_config : mcast_query_interval**: optional string, containing an integer, in range 1 to 3,600
Configures the IP Multicast Querier interval between queries (in seconds). Default: **other_config:mcast_idle_timeout /** 2.

**other_config : mcast_query_max_response**: optional string, containing an integer, in range 1 to 10
Configures the value of the "max-response" field in the multicast queries originated by the logical switch. Default: 1 second.

**other_config : mcast_eth_src**: optional string
Configures the source Ethernet address for queries originated by the logical switch.

**other_config : mcast_ip4_src**: optional string
Configures the source IPv4 address for queries originated by the logical switch.

**other_config : mcast_ip6_src**: optional string
Configures the source IPv6 address for queries originated by the logical switch.

*Interconnection:*

**other_config : interconn-ts**: optional string
The **name** of corresponding transit switch in **OVN_IC_Northbound** database. This kind of logical switch is created and controlled by **ovn−ic**.

**other_config : ic-vxlan_mode**: optional string, either **true** or **false**
**ic−vxlan_mode** is set to true by **ovn−ic** when it runs **VXLAN** as encapsulation protocol for cross-AZ traffic. Default value is false.

*Tunnel Key:*

**other_config : requested-tnl-key**: optional string, containing an integer, in range 1 to 16,777,215
Configures the datapath tunnel key for the logical switch. Usually this is not needed because **ovn−northd** will assign an unique key for each datapath by itself. However, if it is configured, **ovn−northd** honors the configured value. The typical use case is for interconnection: the tunnel keys for transit switches need to be unique globally, so they are maintained in the global **OVN_IC_Southbound** database, and **ovn−ic** simply syncs the value from **OVN_IC_Southbound** through this config.

**copp**: optional weak reference to **Copp**
The control plane protection policy from table **Copp** used for metering packets sent to **ovn−controller** from ports of this logical switch.

*Other options:*

**other_config : vlan-passthru**: optional string, either **true** or **false**
Determines whether VLAN tagged incoming traffic should be allowed. Note that this may have security implications when enabled for a logical switch with a tag=0 localnet port. If not properly isolated from other localnet ports, fabric traffic that belongs to other tagged networks may be passed through such a port.

**other_config : broadcast-arps-to-all-routers**: optional string, either **true** or **false**

Determines whether arp requests and ipv6 neighbor solicitations should be sent to all routers and other switchports (default) or if it should only be sent to switchports where the ip/mac address is unknown. Setting this to false can significantly reduce the load if the logical switch can receive arp requests for ips it does not know about. However setting this to false also means that garps are no longer forwarded to all routers and therefor the mac bindings of the routers are no longer updated.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### Logical_Switch_Port TABLE

A port within an L2 logical switch.

**Summary:**

| | |
|---|---|
| *Core Features:* | |
| **name** | string (must be unique within table) |
| **type** | string |
| *Options:* | |
| **options** | map of string-string pairs |
| *Options for router ports:* | |
| **options : router-port** | optional string |
| **options : nat-addresses** | optional string |
| **options : exclude-lb-vips-from-garp** | optional string |
| **options : exclude-router-ips-from-garp** | optional string |
| **options : arp_proxy** | optional string |
| **options : enable_router_port_acl** | optional string, either **true** or **false** |
| **options : ct-zone-limit** | optional string, containing an integer, in range 0 to 4,294,967,295 |
| **options : dynamic-routing-advertise** | optional string |
| *Options for localnet ports:* | |
| **options : network_name** | optional string |
| **options : ethtype** | optional string |
| **options : localnet_learn_fdb** | optional string, either **true** or **false** |
| *Options for l2gateway ports:* | |
| **options : network_name** | optional string |
| **options : l2gateway-chassis** | optional string |
| *Options for vtep ports:* | |
| **options : vtep-physical-switch** | optional string |
| **options : vtep-logical-switch** | optional string |
| *VMI (or VIF) Options:* | |
| **options : requested-chassis** | optional string |
| **options : requested-encap-ip** | optional string |
| **options : activation-strategy** | optional string |
| **options : iface-id-ver** | optional string |
| **options : qos_min_rate** | optional string |
| **options : qos_max_rate** | optional string |
| **options : qos_burst** | optional string |
| **options : hostname** | optional string |
| **options : force_fdb_lookup** | optional string, either **true** or **false** |
| **options : disable_garp_rarp** | optional string, either **true** or **false** |
| **options : pkt_clone_type** | optional string, must be **mc_unknown** |
| **options : disable_arp_nd_rsp** | optional string, either **true** or **false** |
| **options : lsp_learn_fdb** | optional string, either **true** or **false** |
| **options : receive_multicast** | optional string, either **true** or **false** |
| **options : is-nf** | optional string, either **true** or **false** |
| **options : nf-linked-port** | optional string |
| *VIF Plugging Options:* | |
| **options : vif-plug-type** | optional string |
| **options : vif-plug-mtu-request** | optional string |
| *Virtual port Options:* | |
| **options : virtual-ip** | optional string |
| **options : virtual-parents** | optional string |

| | |
|---|---|
| *IP Multicast Snooping Options:* | |
| **options : mcast_flood** | optional string, either **true** or **false** |
| **options : mcast_flood_reports** | optional string, either **true** or **false** |
| *Containers:* | |
| **parent_name** | optional string |
| **tag_request** | optional integer, in range 0 to 4,095 |
| **tag** | optional integer, in range 1 to 4,095 |
| *Port State:* | |
| **up** | optional boolean |
| **enabled** | optional boolean |
| *Addressing:* | |
| **addresses** | set of strings |
| **dynamic_addresses** | optional string |
| **port_security** | set of strings |
| **peer** | optional string |
| *DHCP:* | |
| **dhcpv4_options** | optional weak reference to **DHCP_Options** |
| **dhcpv6_options** | optional weak reference to **DHCP_Options** |
| **mirror_rules** | set of weak reference to **Mirror**s |
| **health_checks** | set of **Logical_Switch_Port_Health_Check**s |
| **ha_chassis_group** | optional **HA_Chassis_Group** |
| *Naming:* | |
| **external_ids : neutron:port_name** | optional string |
| *Tunnel Key:* | |
| **options : requested-tnl-key** | optional string, containing an integer, in range 1 to 32,767 |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

*Core Features:*

**name**: string (must be unique within table)

The logical port name.

For entities (VMs or containers) that are spawned in the hypervisor, the name used here must match those used in the **external_ids:iface-id** in the **Open_vSwitch** database's **Interface** table, because hypervisors use **external_ids:iface-id** as a lookup key to identify the network interface of that entity.

For containers that share a VIF within a VM, the name can be any unique identifier. See **Containers**, below, for more information.

A logical switch port may not have the same name as a logical router port, but the database schema cannot enforce this.

**type**: string

Specify a type for this logical port. Logical ports can be used to model other types of connectivity into an OVN logical switch. The following types are defined:

(empty string)

A VM (or VIF) interface.

**router**

A connection to a logical router. The value of **options:router-port** specifies the **name** of the **Logical_Router_Port** to which this logical switch port is connected.

**switch**

A connection to another logical switch. The value of **peer** specifies the **name** of the **Logical_Switch_Port** to which this logical switch port is connected. Such ports always have an implicit "unknown" address, because the address information is not leaked between directly connected switches.

**localnet**

A connection to a locally accessible network from **ovn−controller** instances that have a corresponding bridge mapping. A logical switch can have multiple **localnet** ports attached. This type is used to model direct connectivity to existing networks. In this case, each chassis should have a mapping for one of the physical networks only. Note: nothing said above implies that a chassis cannot be plugged to multiple physical networks as long as they belong to different switches.

**localport**

A connection to a local VIF. Traffic that arrives on a **localport** is never forwarded over a tunnel to another chassis. These ports are present on every chassis and have the same address in all of them. This is used to model connectivity to local services that run on every hypervisor.

**l2gateway**

A connection to a physical network.

**vtep** A port to a logical switch on a VTEP gateway.

**external**

Represents a logical port which is external and not having an OVS port in the integration bridge. **OVN** will never receive any traffic from this port or send any traffic to this port. **OVN** can support native services like DHCPv4/DHCPv6/DNS for this port. If **ha_chassis_group** is defined, **ovn−controller** running in the active chassis of the HA chassis group will bind this port to provide these native services. It is expected that this port belong to a bridged logical switch (with a **localnet** port).

It is recommended to use the same HA chassis group for all the external ports of a logical switch. Otherwise, the physical switch might see MAC flap issue when different chassis provide the native services. For example when supporting native DHCPv4 service, DHCPv4 server mac (configured in **options:server_mac** column in table **DHCP_Options**) originating from different ports can cause MAC flap issue. The MAC of the logical router IP(s) can also flap if the same HA chassis group is not set for all the external ports of a logical switch.

Below are some of the use cases where **external** ports can be used.

- VMs connected to SR-IOV nics - Traffic from these VMs by passes the kernel stack and local **ovn−controller** do not bind these ports and cannot serve the native services.
- When CMS supports provisioning baremetal servers.

**virtual**

Represents a logical port which does not have an OVS port in the integration bridge and has a virtual ip configured in the **options:virtual-ip** column. This virtual ip can move around between the logical ports configured in the **options:virtual-parents** column.

One of the use case where **virtual** ports can be used is.

- The **virtual ip** represents a load balancer vip and the **virtual parents** provide load balancer service in an active-standby setup with the active virtual parent owning the **virtual ip**.

**remote**

A remote port is to model a port that resides remotely on another OVN, which is on the other side of a transit logical switch for OVN interconnection. This type of ports are created by **ovn−ic** instead of by CMS. Any change to the port will be automatically overwritten by **ovn−ic**.

*Options:*

**options**: map of string-string pairs

This column provides key/value settings specific to the logical port **type**. The type-specific options are described individually below.

*Options for router ports:*

These options apply when **type** is **router**.

**options : router-port**: optional string

Required. The **name** of the **Logical_Router_Port** to which this logical switch port is connected.

**options : nat-addresses**: optional string

This is used to send gratuitous ARPs for SNAT and DNAT IP addresses via the **localnet** port that is attached to the same logical switch as this type **router** port. This option is specified on a logical switch port that is connected to a gateway router, or a logical switch port that is connected to a distributed gateway port on a logical router.

This must take one of the following forms:

**router**

Gratuitous ARPs will be sent for all SNAT and DNAT external IP addresses and for all load balancer IP addresses defined on the **options:router-port**'s logical router, using the **options:router-port**'s MAC address.

This form of **options:nat-addresses** is valid for logical switch ports where **options:router-port** is the name of a port on a gateway router, or the name of a distributed gateway port.

Supported only in OVN 2.8 and later. Earlier versions required NAT addresses to be manually synchronized.

**Ethernet address followed by one or more IPv4 addresses**

Example: **80:fa:5b:06:72:b7 158.36.44.22 158.36.44.24**. This would result in generation of gratuitous ARPs for IP addresses 158.36.44.22 and 158.36.44.24 with a MAC address of 80:fa:5b:06:72:b7.

This form of **options:nat-addresses** is only valid for logical switch ports where **options:router-port** is the name of a port on a gateway router.

**options : exclude-lb-vips-from-garp**: optional string

If **options:nat-addresses** is set to **router**, Gratuitous ARPs will be sent for all SNAT and DNAT external IP addresses defined on the **options:router-port**'s logical router, using the **options:router-port**'s MAC address, not cosidering configured load balancers.

**options : exclude-router-ips-from-garp**: optional string

If **options:nat-addresses** is set to **router**, Gratuitous ARPs will not be sent for router port IP addresses and SNAT IP addresses (if SNAT IP is same as the router port IP) defined on its peer router port. Do not set this option if the router port IPs are also used as SNAT and DNAT IPs.

**options : arp_proxy**: optional string

Optional. A list of MAC and addresses/cidrs or just addresses/cidrs that this logical switch **router** port will reply to ARP/NDP requests. Examples: **169.254.239.254 169.254.239.2**, **0a:58:a9:fe:01:01 169.254.239.254 169.254.239.2 169.254.238.0/24**, **fd7b:6b4d:7b25:d22f::1 fd7b:6b4d:7b25:d22f::2**, **0a:58:a9:fe:01:01 fd7b:6b4d:7b25:d22f::0/64**. The**options:router-port**'s logical router should have a route to forward packets sent to configured proxy ARP MAC/IPs

to an appropriate destination.

**options : enable_router_port_acl**: optional string, either **true** or **false**
Optional. Enable conntrack for the router port whose peer is l3dgw_port if set to **true**. The default value is **false**.

**options : ct-zone-limit**: optional string, containing an integer, in range 0 to 4,294,967,295
CT zone **limit** value for given **Logical_Switch_Port**. This value has priority over limit specified on **Logical_Switch** when configured. The value 0 means unlimited. When the option is not present the limit is not set and the zone limit is derived from OvS default datapath limit.

**options : dynamic-routing-advertise**: optional string
If the CMS sets **options:dynamic-routing-redistribute** to **connected−as−host**, this parameter is used by the CMS to enable/disable the advertisement of the current host IP via the **Advertised_Route** table. Default: **true**.

*Options for localnet ports:*

These options apply when **type** is **localnet**.

**options : network_name**: optional string
Required. The name of the network to which the **localnet** port is connected. Each hypervisor, via **ovn−controller**, uses its local configuration to determine exactly how to connect to this locally accessible network, if at all.

**options : ethtype**: optional string
Optional. VLAN EtherType field value for encapsulating VLAN headers. Supported values: 802.1q (default), 802.1ad.

**options : localnet_learn_fdb**: optional string, either **true** or **false**
Optional. Allows localnet port to learn MACs and store them in FDB table if set to **true**. The default value is **false**.

*Options for l2gateway ports:*

These options apply when **type** is **l2gateway**.

**options : network_name**: optional string
Required. The name of the network to which the **l2gateway** port is connected. The L2 gateway, via **ovn−controller**, uses its local configuration to determine exactly how to connect to this network.

**options : l2gateway-chassis**: optional string
Required. The chassis on which the **l2gateway** logical port should be bound to. **ovn−controller** running on the defined chassis will connect this logical port to the physical network.

*Options for vtep ports:*

These options apply when **type** is **vtep**.

**options : vtep-physical-switch**: optional string
Required. The name of the VTEP gateway.

**options : vtep-logical-switch**: optional string
Required. A logical switch name connected by the VTEP gateway.

*VMI (or VIF) Options:*

These options apply to logical ports with **type** having (empty string)

**options : requested-chassis**: optional string
If set, identifies a specific chassis (by name or hostname) that is allowed to bind this port. Using this option will prevent thrashing between two chassis trying to bind the same port during a live migration. It can also prevent similar thrashing due to a mis-configuration, if a port is accidentally created on more than one chassis.

If set to a comma separated list, the first entry identifies the main chassis and the rest are one or more additional chassis that are allowed to bind the same port.

When multiple chassis are set for the port, and the logical switch is connected to an external network through a **localnet** port, tunneling is enforced for the port to guarantee delivery of packets directed to the port to all its locations. This has MTU implications because the network used for tunneling must have MTU larger than **localnet** for stable connectivity.

If the same host co-hosts more than one controller instance (either belonging to the same or separate clusters), special attention should be given to consistently using unique chassis names used in this option. It is advised that chassis names - and not host names - are used for this option.

**options : requested-encap-ip**: optional string
Requests the encapsulation IP address for the port binding. If set, **ovn−northd** uses this IP to select the **Encap** entry written to **encap**. This is intended for ports without a local OVS interface, e.g. remote transit switch ports in ovn-kubernetes interconnect mode.

**options : activation-strategy**: optional string
If used with multiple chassis set in **requested-chassis**, specifies an activation strategy for all additional chassis. By default, no activation strategy is used, meaning additional port locations are immediately available for use. The option supports a comma separated list where you can combine 3 protocols, "rarp", "garp" and "na". When any of the protocols is set, the port is blocked for ingress and egress communication until a specified protocol packet is sent from a new location. The activation strategy is useful in live migration scenarios for virtual machines.

**options : iface-id-ver**: optional string
If set, this port will be bound by **ovn−controller** only if this same key and value is configured in the **external_ids** column in the Open_vSwitch database's **Interface** table.

**options : qos_min_rate**: optional string
If set, indicates the minimum guaranteed rate available for data sent from this interface, in bit/s.

**options : qos_max_rate**: optional string
If set, indicates the maximum rate for data sent from this interface, in bit/s. The traffic will be shaped according to this limit.

**options : qos_burst**: optional string
If set, indicates the maximum burst size for data sent from this interface, in bits.

**options : hostname**: optional string
If set, indicates the DHCPv4 option "Hostname" (option code 12) associated for this Logical Switch Port. If DHCPv4 is enabled for this Logical Switch Port, hostname dhcp option will be included in DHCP reply.

**options : force_fdb_lookup**: optional string, either **true** or **false**
This option is supported only if the Logical Switch Port is of default **type** (i.e. type set to empty_string) and also **addresses** column contains **unknown**. If set to **true**, MAC addresses (if configured) are not installed in the l2 lookup table but the MAC addresses are learnt and stored in the FDB table. The default value is **false**.

**options : disable_garp_rarp**: optional string, either **true** or **false**
If set to **true**, GARP and RARP announcements are not sent when a VIF port is created on a bridged logical switch. The default value is **false**.

**options : pkt_clone_type**: optional string, must be **mc_unknown**
If set to mc_unknown, packets going to this VIF get cloned to all unknown ports connected to the same Logical Switch.

**options : disable_arp_nd_rsp**: optional string, either **true** or **false**
If set to **true**, ARP/ND responder flows are not installed for the IP addresses configured on this logical port. Default: **false**.

**options : lsp_learn_fdb**: optional string, either **true** or **false**
If set to **false**, source MAC address is not learnt by FDB when packets are received on this port. The default value is **true**.

**options : receive_multicast**: optional string, either **true** or **false**
If set to **false**, multicast traffic is not forwarded to this port. The default value is **true**.

**options : is-nf**: optional string, either **true** or **false**
Needs to be set to **true** for Network Function ports. These are the ports used as **inport** or **outport** in Network_Function table. The default value is **false**.

**options : nf-linked-port**: optional string
Each row in Network_Function table refers to two logical switch ports under the columns **inport** and **outport**. The port identified as **inport** needs to have this option set to the port identified as **outport**, and vice-versa.

*VIF Plugging Options:*

**options : vif-plug-type**: optional string
If set, OVN will attempt to perform plugging of this VIF. In order to get this port plugged by the OVN controller, OVN must be built with support for VIF plugging. The default behavior is for the CMS to do the VIF plugging. Each VIF plug provider have their own options namespaced by name, for example "vif-plug:representor:key". Please refer to the VIF plug provider documentation located in Documentation/topics/vif-plug-providers/ for more information.

**options : vif-plug-mtu-request**: optional string
Requested MTU for plugged interfaces. When set the OVN controller will fill the **mtu_request** column of the Open vSwitch database's **Interface** table. This in turn will make OVS vswitchd update the MTU of the linked interface.

*Virtual port Options:*

These options apply when **type** is **virtual**.

**options : virtual-ip**: optional string
This option represents the virtual IPv4 address.

**options : virtual-parents**: optional string
This options represents a set of logical port names (with in the same logical switch) which can own the **virtual ip** configured in the **options:virtual-ip**. All these virtual parents should add the **virtual ip** in the **port_security** if port security addressed are enabled.

*IP Multicast Snooping Options:*

These options apply when the port is part of a logical switch which has **other_config** :mcast_snoop set to **true**.

**options : mcast_flood**: optional string, either **true** or **false**
If set to **true**, multicast packets (except reports) are unconditionally forwarded to the specific port. Default: **false**.

**options : mcast_flood_reports**: optional string, either **true** or **false**
If set to **true**, multicast reports are unconditionally forwarded to the specific port. Default: **false**.

*Containers:*

When a large number of containers are nested within a VM, it may be too expensive to dedicate a VIF to each container. OVN can use VLAN tags to support such cases. Each container is assigned a VLAN ID and each packet that passes between the hypervisor and the VM is tagged with the appropriate ID for the container. Such VLAN IDs never appear on a physical wire, even inside a tunnel, so they need not be unique except relative to a single VM on a hypervisor.

These columns are used for VIFs that represent nested containers using shared VIFs. For VMs and for

containers that have dedicated VIFs, they are empty.

**parent_name**: optional string
The VM interface through which the nested container sends its network traffic. This must match the **name** column for some other **Logical_Switch_Port**. Note: for performance of the OVN Southbound database conditional monitoring, unlike for regular VIFs, **ovn−controller** will register to get updates about all OVN Southbound database **Port_Binding** table records that correspond to nested container ports even if **external_ids:ovn−monitor−all** is set to **false**. See **ovn−controller**(8) for more information.

**tag_request**: optional integer, in range 0 to 4,095
The VLAN tag in the network traffic associated with a container's network interface. The client can request **ovn−northd** to allocate a tag that is unique within the scope of a specific parent (specified in **parent_name**) by setting a value of **0** in this column. The allocated value is written by **ovn−northd** in the **tag** column. (Note that these tags are allocated and managed locally in **ovn−northd**, so they cannot be reconstructed in the event that the database is lost.) The client can also request a specific non-zero tag and **ovn−northd** will honor it and copy that value to the **tag** column.

When **type** is set to **localnet** or **l2gateway**, this can be set to indicate that the port represents a connection to a specific VLAN on a locally accessible network. The VLAN ID is used to match incoming traffic and is also added to outgoing traffic.

**tag**: optional integer, in range 1 to 4,095
The VLAN tag allocated by **ovn−northd** based on the contents of the **tag_request** column.

*Port State:*

**up**: optional boolean
This column is populated by **ovn−northd**, rather than by the CMS plugin as is most of this database. When a logical port is bound to a physical location in the OVN Southbound database **Binding** table, **ovn−northd** sets this column to **true**; otherwise, or if the port becomes unbound later, it sets it to **false**. If this column is empty, the port is not considered up. This allows the CMS to wait for a VM's (or container's) networking to become active before it allows the VM (or container) to start.

Logical ports of router type are an exception to this rule. They are considered to be always up, that is this column is always set to **true**.

**enabled**: optional boolean
This column is used to administratively set port state. If this column is empty or is set to **true**, the port is enabled. If this column is set to **false**, the port is disabled. A disabled port has all ingress and egress traffic dropped.

*Addressing:*

**addresses**: set of strings
Addresses owned by the logical port.

Each element in the set must take one of the following forms:

**Ethernet address followed by zero or more IPv4 or IPv6 addresses (or both)**
An Ethernet address defined is owned by the logical port. Like a physical Ethernet NIC, a logical port ordinarily has a single fixed Ethernet address.

When a OVN logical switch processes a unicast Ethernet frame whose destination MAC address is in a logical port's **addresses** column, it delivers it only to that port, as if a MAC learning process had learned that MAC address on the port.

If IPv4 or IPv6 address(es) (or both) are defined, it indicates that the logical port owns the given IP addresses.

If IPv4 address(es) are defined, the OVN logical switch uses this information to synthesize responses to ARP requests without traversing the physical network. The OVN logical router

connected to the logical switch, if any, uses this information to avoid issuing ARP requests for logical switch ports.

Note that the order here is important. The Ethernet address must be listed before the IP address(es) if defined.

Examples:

**80:fa:5b:06:72:b7**
This indicates that the logical port owns the above mac address.

**80:fa:5b:06:72:b7 10.0.0.4 20.0.0.4**
This indicates that the logical port owns the mac address and two IPv4 addresses.

**80:fa:5b:06:72:b7 fdaa:15f2:72cf:0:f816:3eff:fe20:3f41**
This indicates that the logical port owns the mac address and 1 IPv6 address.

**80:fa:5b:06:72:b7 10.0.0.4 fdaa:15f2:72cf:0:f816:3eff:fe20:3f41**
This indicates that the logical port owns the mac address and 1 IPv4 address and 1 IPv6 address.

**unknown**
This indicates that the logical port has an unknown set of Ethernet addresses. When an OVN logical switch processes a unicast Ethernet frame whose destination MAC address is not in any logical port's **addresses** column, it delivers it to the port (or ports) whose **addresses** columns include **unknown**.

When **unknown** is present, FDB learning is enabled for this port: source MACs are recorded via **put_fdb** and destination MACs are resolved via **get_fdb**. For ARP packets, the sender hardware address is also learned. For IPv6 Neighbor Advertisements, the target link-layer address is learned; for Neighbor Solicitations, the source link-layer address is learned.

**dynamic**
Use **dynamic** to make **ovn−northd** generate a globally unique MAC address, choose an unused IPv4 address with the logical port's subnet (if **other_config:subnet** is set in the port's **Logical_Switch**), and generate an IPv6 address from the MAC address (if **other_config:ipv6_prefix** is set in the port's **Logical_Switch**) and store them in the port's **dynamic_addresses** column.

Only one element containing **dynamic** may appear in **addresses**.

**dynamic** *ip*
**dynamic** *ipv6*
**dynamic** *ip ipv6*
These act like **dynamic** alone but specify particular IPv4 or IPv6 addresses to use. OVN IPAM will still automatically allocate the other address if configured appropriately. Example: **dynamic 192.168.0.1 2001::1**.

*mac* **dynamic**
This acts like **dynamic** alone but specifies a particular MAC address to use. OVN IPAM will still automatically allocate IPv4 or IPv6 addresses, or both, if configured appropriately. Example: **80:fa:5b:06:72:b7 dynamic**

**router**
Accepted only when **type** is **router**. This indicates that the Ethernet, IPv4, and IPv6 addresses for this logical switch port should be obtained from the connected logical router port, as specified by **router−port** in **options**.

The resulting addresses are used to populate the logical switch's destination lookup, and also for the logical switch to generate ARP and ND replies.

If the connected logical router port has a distributed gateway port specified and the logical router has rules specified in **nat** with **external_mac**, then those addresses are also used to populate the switch's destination lookup.

Supported only in OVN 2.7 and later. Earlier versions required router addresses to be manually synchronized.

**dynamic_addresses**: optional string

Addresses assigned to the logical port by **ovn−northd**, if **dynamic** is specified in **addresses**. Addresses will be of the same format as those that populate the **addresses** column. Note that dynamically assigned addresses are constructed and managed locally in ovn-northd, so they cannot be reconstructed in the event that the database is lost.

**port_security**: set of strings

This column controls the addresses from which the host attached to the logical port (''the host'') is allowed to send packets and to which it is allowed to receive packets. If this column is empty, all addresses are permitted.

Each element in the set must begin with one Ethernet address. This would restrict the host to sending packets from and receiving packets to the ethernet addresses defined in the logical port's **port_security** column. It also restricts the inner source MAC addresses that the host may send in ARP and IPv6 Neighbor Discovery packets. The host is always allowed to receive packets to multicast and broadcast Ethernet addresses.

Each element in the set may additionally contain one or more IPv4 or IPv6 addresses (or both), with optional masks. If a mask is given, it must be a CIDR mask. In addition to the restrictions described for Ethernet addresses above, such an element restricts the IPv4 or IPv6 addresses from which the host may send and to which it may receive packets to the specified addresses. A masked address, if the host part is zero, indicates that the host is allowed to use any address in the subnet; if the host part is nonzero, the mask simply indicates the size of the subnet. In addition:

- If any IPv4 address is given, the host is also allowed to receive packets to the IPv4 local broadcast address 255.255.255.255 and to IPv4 multicast addresses (224.0.0.0/4). If an IPv4 address with a mask is given, the host is also allowed to receive packets to the broadcast address in that specified subnet.

  If any IPv4 address is given, the host is additionally restricted to sending ARP packets with the specified source IPv4 address. (RARP is not restricted.)

- If any IPv6 address is given, the host is also allowed to receive packets to IPv6 multicast addresses (ff00::/8).

  If any IPv6 address is given, the host is additionally restricted to sending IPv6 Neighbor Discovery Solicitation or Advertisement packets with the specified source address or, for solicitations, the unspecified address.

If an element includes an IPv4 address, but no IPv6 addresses, then IPv6 traffic is not allowed. If an element includes an IPv6 address, but no IPv4 address, then IPv4 and ARP traffic is not allowed.

This column uses the same lexical syntax as the **match** column in the OVN Southbound database's **Pipeline** table. Multiple addresses within an element may be space or comma separated.

Each element in the set also supports the special prefix "VRRPv3" that allows specification of single physical MAC and multiple VRRPv3 MAC addresses. As for non VRRPv3 entries, multiple IP addresses can be associated with the specified MACs. "VRRPv3" with a single physical MAC translates to allowing traffic for the whole "VRRPv3" range of MACs. See more in the examples. When the port security configuration entry contains the VRRPv3 label, the behavior for physical MAC (the first specified) is different. The installed flows will allow traffic from/to VRRP MACs + IPs. The physical MAC is there to properly allow ARP/ND with given VRRP MACs. To allow traffic that is not related to virtual router, e.g., IP traffic with a physical MAC as a source, a regular port security entry should

be added separately.

This column is provided as a convenience to cloud management systems, but all of the features that it implements can be implemented as ACLs using the **ACL** table.

Examples:

**80:fa:5b:06:72:b7**
: The host may send traffic from and receive traffic to the specified MAC address, and to receive traffic to Ethernet multicast and broadcast addresses, but not otherwise. The host may not send ARP or IPv6 Neighbor Discovery packets with inner source Ethernet addresses other than the one specified.

**80:fa:5b:06:72:b7 192.168.1.10/24**
: This adds further restrictions to the first example. The host may send IPv4 packets from or receive IPv4 packets to only 192.168.1.10, except that it may also receive IPv4 packets to 192.168.1.255 (based on the subnet mask), 255.255.255.255, and any address in 224.0.0.0/4. The host may not send ARPs with a source Ethernet address other than 80:fa:5b:06:72:b7 or source IPv4 address other than 192.168.1.10. The host may not send or receive any IPv6 (including IPv6 Neighbor Discovery) traffic.

**"80:fa:5b:12:42:ba", "80:fa:5b:06:72:b7 192.168.1.10/24"**
: The host may send traffic from and receive traffic to the specified MAC addresses, and to receive traffic to Ethernet multicast and broadcast addresses, but not otherwise. With MAC 80:fa:5b:12:42:ba, the host may send traffic from and receive traffic to any L3 address. With MAC 80:fa:5b:06:72:b7, the host may send IPv4 packets from or receive IPv4 packets to only 192.168.1.10, except that it may also receive IPv4 packets to 192.168.1.255 (based on the subnet mask), 255.255.255.255, and any address in 224.0.0.0/4. The host may not send or receive any IPv6 (including IPv6 Neighbor Discovery) traffic.

**"VRRPv3 <PHYSICAL_MAC>"**
: The host may send ARP/ND packets with physical MAC as a source and any of the VRRPv3 MACs as inner SHA/TLL/SLL - 00:00:5e:00:01:00/40 for IPv4 and 00:00:5e:00:02:00/40 for IPv6. It may also send or receive any traffic with any of VRRPv3 MAC addresses as a source or destination. But not with the physical MAC address.

**"VRRPv3 <PHYSICAL_MAC> <VRRPV3_MACv4_1>/[<MASK1>] <VRRPV3_MACv4_N> <VRRPV3_MACv6_1>/[<MASK2>] <VRRPV3_MACv6_N>"**
: Same as the previous example, but allowed VRRPv3 MAC addresses are limited to the specified ones. The specified VRRPv3 MAC addresses must have a correct prefix - 00:00:5e:00:01 for IPv4 and 00:00:5e:00:02 for IPv6. VRRPv3 MACs can be provided with a mask with a prefix between /40 and /48.

**"VRRPv3 <PHYSICAL_MAC> <VRRPV3_MACv4_1>/[<MASK1>] <VRRPV3_MACv4_N> <VRRPV3_MACv6_1>/[<MASK2>] <VRRPV3_MACv6_N> <VRRPV3_IPv4_1>/[<MASK1>] <VRRPV3_IPv4_N> <VRRPV3_IPv6_1>/[<MASK2>] <VRRPV3_IPv6_N>"**
: The same as the previous example, but the provided IP addresses further restrict the inner IP for ARP/ND. As well as IP/IPv6 traffic to/from given VRRPv3 MACs. The IP address format is the same as for the regular port security entry.

**peer**: optional string
: For a switch port used to connect two logical switches, this identifies the other switch port in the pair by **name**.

  For a switch port attached to a logical router, this column is empty.

*DHCP:*

**dhcpv4_options**: optional weak reference to **DHCP_Options**
This column defines the DHCPv4 Options to be included by the **ovn−controller** when it replies to the DHCPv4 requests. Please see the **DHCP_Options** table.

**dhcpv6_options**: optional weak reference to **DHCP_Options**
This column defines the DHCPv6 Options to be included by the **ovn−controller** when it replies to the DHCPv6 requests. Please see the **DHCP_Options** table.

**mirror_rules**: set of weak reference to **Mirror**s
Mirror rules that apply to logical switch port which is the source. Please see the **Mirror** table.

**health_checks**: set of **Logical_Switch_Port_Health_Check**s
Health checks configuration for this logical switch port. Please see the **Logical_Switch_Port_Health_Check** table.

**ha_chassis_group**: optional **HA_Chassis_Group**
References a row in the OVN Northbound database's **HA_Chassis_Group** table. It indicates the HA chassis group to use if the **type** is set to **external**. If **type** is not **external**, this column is ignored.

*Naming:*

**external_ids : neutron:port_name**: optional string
This column gives an optional human-friendly name for the port. This name has no special meaning or purpose other than to provide convenience for human interaction with the northbound database.

Neutron copies this from its own port object's name. (Neutron ports do are not assigned human-friendly names by default, so it will often be empty.)

*Tunnel Key:*

**options : requested-tnl-key**: optional string, containing an integer, in range 1 to 32,767
Configures the port binding tunnel key for the port. Usually this is not needed because **ovn−northd** will assign an unique key for each port by itself. However, if it is configured, **ovn−northd** honors the configured value. The typical use case is for interconnection: the tunnel keys for ports on transit switches need to be unique globally, so they are maintained in the global **OVN_IC_Southbound** database, and **ovn−ic** simply syncs the value from **OVN_IC_Southbound** through this config.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

The **ovn−northd** program copies all these pairs into the **external_ids** column of the **Port_Binding** table in **OVN_Southbound** database.

### Logical_Switch_Port_Health_Check TABLE

Each row represents health check configuration for logical switch port. Health checks are used to monitor reachability and health of backend endpoints associated with logical switch port. Health check can be configured to use different protocols (TCP, UDP, or ICMP) to verify availability of target IP addresses.

**Summary:**

| | |
|---|---|
| **protocol** | string, one of **icmp**, **tcp**, or **udp** |
| **src_ip** | string |
| **port** | integer, in range 0 to 65,535 |
| **address** | string |
| **status** | optional string, one of **error**, **offline**, or **online** |
| **options : interval** | optional string, containing an integer |
| **options : timeout** | optional string, containing an integer |
| **options : success_count** | optional string, containing an integer |
| **options : failure_count** | optional string, containing an integer |

**Details:**

**protocol**: string, one of **icmp**, **tcp**, or **udp**

Valid protocols are **tcp**, **udp**, or **icmp**. For **tcp** and **udp** protocols - destination port must be specified.

**src_ip**: string

Source IP address used when sending health check probes.

**port**: integer, in range 0 to 65,535

Destination port number for TCP and UDP health checks.

**address**: string

IP address to monitor for the health check.

**status**: optional string, one of **error**, **offline**, or **online**

Health status synchronized from the **status** field of corresponding service monitor in the SBDB. This status indicates the current availability of logical switch port for the CMS to determine service virtual machine health and make routing or failover decisions.

**options : interval**: optional string, containing an integer

The interval, in seconds, between service monitor checks.

**options : timeout**: optional string, containing an integer

The time, in seconds, after which the service monitor check times out.

**options : success_count**: optional string, containing an integer

The number of successful checks after which the service is considered **online**.

**options : failure_count**: optional string, containing an integer

The number of failure checks after which the service is considered **offline**.

### Forwarding_Group TABLE

Each row represents one forwarding group.

**Summary:**

| | |
|---|---|
| **name** | string |
| **vip** | string |
| **vmac** | string |
| **liveness** | boolean |
| **child_port** | set of 1 or more strings |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string
: A name for the forwarding group. This name has no special meaning or purpose other than to provide convenience for human interaction with the ovn-nb database.

**vip**: string
: The virtual IP address assigned to the forwarding group. It will respond with vmac when an ARP request is sent for vip.

**vmac**: string
: The virtual MAC address assigned to the forwarding group.

**liveness**: boolean
: If set to **true**, liveness is enabled for child ports otherwise it is disabled.

**child_port**: set of 1 or more strings
: List of child ports in the forwarding group.

*Common Columns:*

**external_ids**: map of string-string pairs
: See **External IDs** at the beginning of this document.

### Address_Set TABLE

Each row in this table represents a named set of addresses. An address set may contain Ethernet, IPv4, or IPv6 addresses with optional bitwise or CIDR masks. Address set may ultimately be used in ACLs to compare against fields such as **ip4.src** or **ip6.src**. A single address set must contain addresses of the same type. As an example, the following would create an address set with three IP addresses:

**ovn−nbctl create Address_Set name=set1 addresses='10.0.0.1 10.0.0.2 10.0.0.3'**

Address sets may be used in the **match** column of the **ACL** table. For syntax information, see the details of the expression language used for the **match** column in the **Logical_Flow** table of the **OVN_Southbound** database.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **addresses** | set of strings |
| **options** | map of string-string pairs |
| **options : ic-adv** | optional string |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)
A name for the address set. Names are ASCII and must match **[a−zA−Z_.][a−zA−Z_.0−9]***.

**addresses**: set of strings
The set of addresses in string form.

**options**: map of string-string pairs
This column provides general key/value settings. The supported options are described individually below.

**options : ic-adv**: optional string
A boolean value that enables advertisement of this Address_Set to the global OVN_IC_Southbound database. Default is **false**.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

## Port_Group TABLE

Each row in this table represents a named group of logical switch ports.

Port groups may be used in the **match** column of the **ACL** table. For syntax information, see the details of the expression language used for the **match** column in the **Logical_Flow** table of the **OVN_Southbound** database.

For each port group, there are two address sets generated to the **Address_Set** table of the **OVN_Southbound** database, containing the IP addresses of the group of ports, one for IPv4, and the other for IPv6, with **name** being the **name** of the **Port_Group** followed by a suffix **_ip4** for IPv4 and **_ip6** for IPv6. The generated address sets can be used in the same way as regular address sets in the **match** column of the **ACL** table. For syntax information, see the details of the expression language used for the **match** column in the **Logical_Flow** table of the **OVN_Southbound** database.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **ports** | set of weak reference to **Logical_Switch_Port**s |
| **acls** | set of **ACL**s |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)

A name for the port group. Names are ASCII and must match **[a−zA−Z_.][a−zA−Z_.0−9]***.

**ports**: set of weak reference to **Logical_Switch_Port**s

The logical switch ports belonging to the group in uuids.

**acls**: set of **ACL**s

Access control rules that apply to the port group. Applying an ACL to a port group has the same effect as applying the ACL to all logical lswitches that the ports of the port group belong to.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### Load_Balancer TABLE

Each row represents one load balancer.

**Summary:**

| | |
|---|---|
| **name** | string |
| **vips** | map of string-string pairs |
| **protocol** | optional string, one of **sctp**, **tcp**, or **udp** |
| *Health Checks:* | |
| **health_check** | set of **Load_Balancer_Health_Check**s |
| **ip_port_mappings** | map of string-string pairs |
| **selection_fields** | set of strings, one of **eth_dst**, **eth_src**, **ip_dst**, **ip_src**, **ipv6_dst**, **ipv6_src**, **tp_dst**, or **tp_src** |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |
| *Load_Balancer options:* | |
| **options : reject** | optional string, either **true** or **false** |
| **options : hairpin_snat_ip** | optional string |
| **options : skip_snat** | optional string |
| **options : add_route** | optional string |
| **options : neighbor_responder** | optional string |
| **options : template** | optional string |
| **options : address-family** | optional string |
| **options : affinity_timeout** | optional string |
| **options : ct_flush** | optional string, either **true** or **false** |
| **options : use_stateless_nat** | optional string, either **true** or **false** |
| **options : distributed** | optional string |
| **options : event** | optional string, either **true** or **false** |
| **options : dynamic-routing-advertise** | optional string |

**Details:**

**name**: string

A name for the load balancer. This name has no special meaning or purpose other than to provide convenience for human interaction with the ovn-nb database.

**vips**: map of string-string pairs

A map of virtual IP addresses (and an optional port number with **:** as a separator) associated with this load balancer and their corresponding endpoint IP addresses (and optional port numbers with **:** as separators) separated by commas. If the destination IP address (and port number) of a packet leaving a container or a VM matches the virtual IP address (and port number) provided here as a key, then OVN will statefully replace the destination IP address by one of the provided IP address (and port number) in this map as a value. IPv4 and IPv6 addresses are supported for load balancing; however a VIP of one address family may not be mapped to a destination IP address of a different family. If specifying an IPv6 address with a port, the address portion must be enclosed in square brackets. Examples for keys are "192.168.1.4" and "[fd0f::1]:8800". Examples for value are "10.0.0.1, 10.0.0.2" and "20.0.0.10:8800, 20.0.0.11:8800".

When the **Load_Balancer** is added to the **logical_switch**, the VIP has to be in a different subnet than the one used for the **logical_switch**. Since VIP is in a different subnet, you should connect your logical switch to either a OVN logical router or a real router (this is because the client can now send a packet with VIP as the destination IP address and router's mac address as the destination MAC address).

**protocol**: optional string, one of **sctp**, **tcp**, or **udp**

Valid protocols are **tcp**, **udp**, or **sctp**. This column is useful when a port number is provided as part of the **vips** column. If this column is empty and a port number is provided as part of **vips** column, OVN assumes the protocol to be **tcp**.

*Health Checks:*

OVN supports health checks for load balancer endpoints. When health checks are enabled, the load balancer uses only healthy endpoints.

Suppose that **vips** contains a key-value pair **10.0.0.10:80=10.0.0.4:8080,20.0.0.4:8080**. To enable health checks for this virtual's endpoints, add two key-value pairs to **ip_port_mappings**, with keys **10.0.0.4** and **20.0.0.4**, and add to **health_check** a reference to a **Load_Balancer_Health_Check** row whose **vip** is set to **10.0.0.10**. The same approach can be used for IPv6 as well.

**health_check**: set of **Load_Balancer_Health_Check**s
Load balancer health checks associated with this load balancer.

**ip_port_mappings**: map of string-string pairs
Maps from endpoint IP to a colon-separated pair of logical port name and source IP, e.g. *port_name***:***source_ip* for IPv4. Health checks are sent to this port with the specified source IP. For IPv6 square brackets must be used around IP address, e.g: *port_name***:***[source_ip]*. The source IP must be from the subnet of the monitored endpoint. It can be either an unused IP from the subnet, or an IP of one of the Logical Router Ports connected to the same switch. Remote endpoint: Specify :target_zone_name at the end of the above syntax to create remote health checks in a specific zone. For distributed load balancers - ip_port_mappings is required. In the absence of health checks - source_ip is optional.

For example, in the example above, IP to port mappings might be defined as **10.0.0.4=sw0−p1:10.0.0.2** and **20.0.0.4=sw1−p1:20.0.0.2**, if the values given were suitable ports and IP addresses. And remote endpoint: **10.0.0.4=sw0−p1:10.0.0.2**:az1, where **sw0−p1** - logical port in **az1**.

For IPv6 IP to port mappings might be defined as **[2001::1]=sw0−p1:[2002::1]**. Remote endpoint: same as for IP.

**selection_fields**: set of strings, one of **eth_dst**, **eth_src**, **ip_dst**, **ip_src**, **ipv6_dst**, **ipv6_src**, **tp_dst**, or **tp_src**
OVN native load balancers are supported using the OpenFlow groups of type **select**. OVS supports two selection methods: **dp_hash** and **hash (with optional fields specified)** in selecting the buckets of a group. Please see the OVS documentation (man ovs-ofctl) for more details on the selection methods. Each endpoint IP (and port if set) is mapped to a bucket in the group flow.

CMS can choose the **hash** selection method by setting the selection fields in this column. **ovs−vswitchd** uses the specified fields in generating the hash.

Example: **{ip_proto,ip_src,ip_dst}** for a 3-tuple match. Example: **{ip_proto,ipv6_src,ipv6_dst}** for an IPv6 match. Example: **{ip_proto,ip_src,ip_dst,tp_src,tp_dst}**. Example: **{ip_src,ip_dst,ipv6_src,ipv6_dst,tp_src,tp_dst}**.

**dp_hash** selection method uses the assistance of datapath to calculate the hash and it is expected to be faster than **hash** selection method. So CMS should take this into consideration before using the **hash** method. Please consult the OVS documentation and OVS sources for the implementation details.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

*Load_Balancer options:*

**options : reject**: optional string, either **true** or **false**
If the load balancer is created with **−−reject** option and it has no active backends, a TCP reset segment (for tcp) or an ICMP port unreachable packet (for all other kind of traffic) will be sent whenever an incoming packet is received for this load-balancer. Please note using **−−reject** option will disable empty_lb SB controller event for this load balancer.

**options : hairpin_snat_ip**: optional string
IP to be used as source IP for packets that have been hair-pinned after load balancing. The default behavior when the option is not set is to use the load balancer VIP as source IP. This option may have exactly one IPv4 and/or one IPv6 address on it, separated by a space character.

**options : skip_snat**: optional string
If the load balancing rule is configured with **skip_snat** option, the option lb_force_snat_ip configured for the logical router that references this load balancer will not be applied for this load balancer.

**options : add_route**: optional string
If set to **true**, then neighbor routers will have logical flows added that will allow for routing to the VIP IP. It also will have ARP resolution logical flows added. By setting this option, it means there is no reason to create a **Logical_Router_Static_Route** from neighbor routers to this NAT address. It also means that no ARP request is required for neighbor routers to learn the IP-MAC mapping for this VIP IP. For more information about what flows are added for IP routes, please see the **ovn−northd** manpage section on IP Routing.

**options : neighbor_responder**: optional string
If set to **all**, then routers on which the load balancer is applied reply to ARP/neighbor discovery requests for all VIPs of the load balancer. If set to **reachable**, then routers on which the load balancer is applied reply to ARP/neighbor discovery requests only for VIPs that are part of a router's subnet. If set to **none**, then routers on which the load balancer is applied never reply to ARP/neighbor discovery requests for any of the load balancer VIPs. Load balancers with **options:template=true** do not support **reachable** as a valid mode. The default value of this option, if not specified, is **reachable** for regular load balancers and **none** for template load balancers.

**options : template**: optional string
Option to be set to **true**, if the load balancer is a template. The load balancer VIPs and backends must be using **Chassis_Template_Var** in their definitions.

Load balancer template VIP supported formats are:

**^VIP_VAR[:^PORT_VAR|:port]**

where **VIP_VAR** and **PORT_VAR** are keys of the **Chassis_Template_Var variables** records.

Note: The VIP and PORT cannot be combined into a single template variable. For example, a **Chassis_Template_Var** variable expanding to **10.0.0.1:8080** is not valid if used as VIP.

Load balancer template backend supported formats are:

**^BACKEND_VAR1[:^PORT_VAR1|:port],^BACKEND_VAR2[:^PORT_VAR2|:port]**
**or**
**^BACKENDS_VAR1,^BACKENDS_VAR2**

where **BACKEND_VAR1**, **PORT_VAR1**, **BACKEND_VAR2**, **PORT_VAR2**, **BACKENDS_VAR1** and **BACKENDS_VAR2** are keys of the **Chassis_Template_Var variables** records.

In the above examples, square brackets are only used to indicate choosing between the enclosed options. But when adding to **Chassis_Template_Var variables** backend IPv6 addresses must be enclosed in []. VIPs must not be enclosed in []. For example: **lbvip="3001::10",lbport=12010,lbback="[2001::1]", ip6_back="[2001::1]:12010"**

**options : address-family**: optional string
Address family used by the load balancer. Supported values are **ipv4** and **ipv6**. The address-family is only used for load balancers with **options:template=true**. For explicit load balancers, setting the address-family has no effect.

**options : affinity_timeout**: optional string
If the CMS provides a positive value (in seconds) for **affinity_timeout**, OVN will dnat connections received from the same client to this lb to the same backend if received in the affinity timeslot. Max supported affinity_timeout is 65535 seconds.

**options : ct_flush**: optional string, either **true** or **false**
The value indicates whether ovn-controller should flush CT entries that are related to this LB. The flush happens if the LB is removed, any of the backends is updated/removed or the LB is not considered local anymore by the ovn-controller. This option is set to **false** by default.

**options : use_stateless_nat**: optional string, either **true** or **false**
If the load balancer is configured with **use_stateless_nat** option to **true**, the logical router that references this load balancer will use Stateless NAT rules when the logical router has multiple distributed gateway ports(DGP). Otherwise, the outbound traffic may be dropped in scenarios where we have different chassis for each DGP. This option is set to **false** by default.

**options : distributed**: optional string
Option enables distributed load balancing across compute nodes, ensuring traffic is always routed to local backends, eliminating east-west traffic between nodes. Required configuration: **ip_port_mappings**. NOTE: The addressing of the underlay network must not overlap with the addressing of Load Balancer VIP. If the Load Balancer is attached to a router that is directly connected to the underlay network and the VIP belongs to the same subnet as used on the underlay network, the traffic won't be spread across all chassis. Instead, it will be concentrated only on the chassis that hosts the Distributed Gateway Port of the router.

**options : event**: optional string, either **true** or **false**
If set to **true** and the load balancer has no active backends, whenever a packet matches a VIP on this load balancer, an **empty_lb_backends** event is reported in the **Controller_Event** table in the OVN Southbound database. This is the per-load-balancer replacement for the deprecated global **controller_event** option in the **NB_Global** table. It is possible to associate a meter named **event−elb** to rate-limit these events. The default value is **false**. If both **event** and **reject** are set, **reject** takes precedence and no event is generated.

**options : dynamic-routing-advertise**: optional string
If the CMS sets **options:dynamic-routing-redistribute** to **lb**, this parameter is used by the CMS to enable/disable the advertisement of the current LB Service IP via the **Advertised_Route** table. Default: **true**.

### Load_Balancer_Group TABLE

Each row represents a logical grouping of load balancers. It is up to the CMS to decide the criteria on which load balancers are grouped together. To simplify configuration and to optimize its processing load balancers that must be associated to the same set of logical switches and/or logical routers should be grouped together.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **load_balancer** | set of weak reference to **Load_Balancer**s |

**Details:**

**name**: string (must be unique within table)

A name for the load balancer group. This name has no special meaning or purpose other than to provide convenience for human interaction with the ovn-nb database.

**load_balancer**: set of weak reference to **Load_Balancer**s

A set of load balancers.

### Load_Balancer_Health_Check TABLE

Each row represents one load balancer health check.

**Summary:**

| | |
|---|---|
| **vip** | string |
| *Health check options:* | |
| **options : interval** | optional string, containing an integer |
| **options : timeout** | optional string, containing an integer |
| **options : success_count** | optional string, containing an integer |
| **options : failure_count** | optional string, containing an integer |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**vip**: string

**vip** whose endpoints should be monitored for health check.

*Health check options:*

**options : interval**: optional string, containing an integer

The interval, in seconds, between health checks.

**options : timeout**: optional string, containing an integer

The time, in seconds, after which a health check times out.

**options : success_count**: optional string, containing an integer

The number of successful checks after which the endpoint is considered online.

**options : failure_count**: optional string, containing an integer

The number of failure checks after which the endpoint is considered offline.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

## ACL TABLE

Each row in this table represents one ACL rule for a logical switch or a port group that points to it through its **acls** column. The **action** column for the highest-**priority** matching row in this table determines a packet's treatment. If no row matches, packets are allowed by default. (Default-deny treatment is possible: add a rule with **priority** 0, **1** as **match**, and **deny** as **action**.)

**Summary:**

| | |
|---|---|
| **label** | integer, in range 0 to 4,294,967,295 |
| **priority** | integer, in range 0 to 32,767 |
| **direction** | string, either **from−lport** or **to−lport** |
| **match** | string |
| **action** | string, one of **allow−related**, **allow−stateless**, **allow**, **drop**, **pass**, or **reject** |
| **tier** | integer, in range 0 to 3 |
| **network_function_group** | optional **Network_Function_Group** |
| *options:* | |
| **options : apply-after-lb** | optional string |
| **options : persist-established** | optional string |
| *Logging:* | |
| **log** | boolean |
| **name** | optional string, at most 63 characters long |
| **severity** | optional string, one of **alert**, **debug**, **info**, **notice**, or **warning** |
| **meter** | optional string |
| **sample_new** | optional **Sample** |
| **sample_est** | optional **Sample** |
| *Common Columns:* | |
| **options** | map of string-string pairs |
| *ACL configuration options:* | |
| **options : log-related** | optional string |
| **external_ids** | map of string-string pairs |

**Details:**

**label**: integer, in range 0 to 4,294,967,295

Associates an identifier with the ACL. The same value will be written to corresponding connection tracker entry. The value should be a valid 32-bit unsigned integer. This value can help in debugging from connection tracker side. For example, through this "label" we can backtrack to the ACL rule which is causing a "leaked" connection. Connection tracker entries are created only for allowed connections so the label is valid only for allow and allow-related actions.

Note: if an ACL has both sampling enabled and a label associated to it then the label value overrides the observation point ID defined in the **sample_new** or **sample_est** configuration.

**priority**: integer, in range 0 to 32,767

The ACL rule's priority. Rules with numerically higher priority take precedence over those with lower. If two ACL rules with the same priority both match, then the one actually applied to a packet is undefined.

Return traffic from an **allow−related** flow is always allowed and cannot be changed through an ACL.

**allow−stateless** flows always take precedence before stateful ACLs, regardless of their priority. (Both **allow** and **allow−related** ACLs can be stateful.)

**direction**: string, either **from−lport** or **to−lport**

Direction of the traffic to which this rule should apply:

- **from−lport**: Used to implement filters on traffic arriving from a logical port. These rules are applied to the logical switch's ingress pipeline.
- **to−lport**: Used to implement filters on traffic forwarded to a logical port. These rules are applied to the logical switch's egress pipeline.

**match**: string
The packets that the ACL should match, in the same expression language used for the **match** column in the OVN Southbound database's **Logical_Flow** table. The **outport** logical port is only available in the **to−lport** direction (the **inport** is available in both directions).

By default all traffic is allowed. When writing a more restrictive policy, it is important to remember to allow flows such as ARP and IPv6 neighbor discovery packets.

Note that you can not create an ACL matching on a port with type=router or type=localnet.

**action**: string, one of **allow−related**, **allow−stateless**, **allow**, **drop**, **pass**, or **reject**
The action to take when the ACL rule matches:

- **allow−stateless**: Always forward the packet in stateless manner, omitting connection tracking mechanism, regardless of other rules defined for the switch. May require defining additional rules for inbound replies. For example, if you define a rule to allow outgoing TCP traffic directed to an IP address, then you probably also want to define another rule to allow incoming TCP traffic coming from this same IP address. In addition, traffic that matches stateless ACLs will bypass load-balancer DNAT/un-DNAT processing. Stateful ACLs should be used instead if the traffic is supposed to be load-balanced.
- **allow**: Forward the packet. It will also send the packets through connection tracking when **allow−related** rules exist on the logical switch. Otherwise, it's equivalent to **allow−stateless**.
- **allow−related**: Forward the packet and related traffic (e.g. inbound replies to an outbound connection).
- **drop**: Silently drop the packet.
- **reject**: Drop the packet, replying with a RST for TCP or ICMPv4/ICMPv6 unreachable message for other IPv4/IPv6-based protocols.
- **pass**: Pass to the next ACL tier. If using multiple ACL tiers, a match on this ACL will stop evaluating ACLs at the current tier and move to the next one. If not using ACL tiers or if a **pass** ACL is matched at the final tier, then the **options:default_acl_drop** option from the **NB_Global** table is used to determine how to proceed.

**tier**: integer, in range 0 to 3
The hierarchical tier that this ACL belongs to.

ACLs can be assigned to numerical tiers. When evaluating ACLs, an internal counter is used to determine which tier of ACLs should be evaluated. Tier 0 ACLs are evaluated first. If no verdict can be determined, then tier 1 ACLs are evaluated next. This continues until the maximum tier value is reached. If all tiers of ACLs are evaluated and no verdict is reached, then the **options:default_acl_drop** option from table **NB_Global** is used to determine how to proceed.

In this version of OVN, the maximum tier value for ACLs is 3, meaning there are 4 tiers of ACLs allowed (0−3).

**network_function_group**: optional **Network_Function_Group**
Group of network functions to which the traffic matching this ACL is redirected.

*options:*

ACLs options.

**options : apply-after-lb**: optional string
If set to true, the ACL will be applied after load balancing stage. Supported only for **from−lport** direction.

The main use case of this option is to support ACLs matching on the destination IP address of the packet for the backend IPs of load balancers.

**OVN** will apply the **from−lport** ACLs in two stages. ACLs without this option **apply−after−lb** set, will be applied before the load balancer stage and ACLs with this option set will be applied after the load balancer stage. The priorities are indepedent between these stages and may not be obvious to the CMS. Hence CMS should be extra careful when using this option and should carefully evaluate the priorities of all the ACLs and the default deny/allow ACLs if any.

**options : persist-established**: optional string
This option applies only to ACLs whose **action** is set to **allow−related**.

**allow−related** ACLs create a conntrack entry when a packet matches the ACL's **match** column. Typically, traffic must continue to match these conditions in order to continue to be allowed by the ACL. With this option set to **true**, then the ACL match is bypassed once the original match occurs. Instead, a mark bit in the conntrack entry is used to allow the traffic. This means that traffic will continue to be allowed even if the ACL's match changes and no longer matches the established traffic.

The traffic will stop being allowed automatically if this option is set to **false**, if the ACL's **action** is changed to something other than **allow−related**, or if the ACL is destroyed.

*Logging:*

These columns control whether and how OVN logs packets that match an ACL.

**log**: boolean
If set to **true**, packets that match the ACL will trigger a log message on the transport node or nodes that perform ACL processing. Logging may be combined with any **action**.

If set to **false**, the remaining columns in this group have no significance.

**name**: optional string, at most 63 characters long
This name, if it is provided, is included in log records. It provides the administrator and the cloud management system a way to associate a log record with a particular ACL.

**severity**: optional string, one of **alert**, **debug**, **info**, **notice**, or **warning**
The severity of the ACL. The severity levels match those of syslog, in decreasing level of severity: **alert**, **warning**, **notice**, **info**, or **debug**. When the column is empty, the default is **info**.

**meter**: optional string
The name of a meter to rate-limit log messages for the ACL. The string must match the **name** column of a row in the **Meter** table. By default, log messages are not rate-limited. In order to ensure that the same **Meter** rate limits multiple ACL logs separately, set the **fair** column.

**sample_new**: optional **Sample**
The entry in the **Sample** table to use for sampling for new sessions matched by this ACL. In case the ACL is stateless this is used for sampling all traffic matched by the ACL.

Note: if an ACL has both sampling enabled and a label associated to it then the label value overrides the observation point ID defined in the **sample_new** configuration.

**sample_est**: optional **Sample**
The entry in the **Sample** table to use for sampling for established/related sessions matched by this ACL.

Note: if an ACL has both sampling enabled and a label associated to it then the label value overrides the observation point ID defined in the **sample_est** configuration.

*Common Columns:*

**options**: map of string-string pairs
This column provides general key/value settings. The supported options are described individually below.

*ACL configuration options:*

**options : log-related**: optional string
If set to **true**, then log when reply or related traffic is admitted from a stateful ACL. In order for this option to function, the **log** option must be set to **true** and a **label** must be set, and it must be unique to the ACL. The label is necessary as it is the only means to associate the reply traffic with the ACL to which it belongs. It must be unique, because otherwise it is ambiguous which ACL will be matched. Note: If this option is enabled, an extra flow is installed in order to log the related traffic. Therefore, if this is enabled on all ACLs, then the total number of flows necessary to log the ACL traffic is doubled, compared to if this option is not enabled.

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

### Logical_Router TABLE

Each row represents one L3 logical router.

**Summary:**

| | |
|---|---|
| **ports** | set of **Logical_Router_Port**s |
| **static_routes** | set of **Logical_Router_Static_Route**s |
| **policies** | set of **Logical_Router_Policy**s |
| **enabled** | optional boolean |
| **nat** | set of **NAT**s |
| **load_balancer** | set of weak reference to **Load_Balancer**s |
| **load_balancer_group** | set of **Load_Balancer_Group**s |
| *Naming:* | |
| **name** | string |
| **external_ids : neutron:router_name** | optional string |
| **copp** | optional weak reference to **Copp** |
| *Options:* | |
| **options : chassis** | optional string |
| **options : dnat_force_snat_ip** | optional string |
| **options : lb_force_snat_ip** | optional string |
| **options : mcast_relay** | optional string, either **true** or **false** |
| **options : dynamic_neigh_routers** | optional string, either **true** or **false** |
| **options : always_learn_from_arp_request** | optional string, either **true** or **false** |
| **options : requested-tnl-key** | optional string, containing an integer, in range 1 to 16,777,215 |
| **options : snat-ct-zone** | optional string, containing an integer, in range 0 to 65,535 |
| **options : mac_binding_age_threshold** | optional string |
| **options : ct-zone-limit** | optional string, containing an integer, in range 0 to 4,294,967,295 |
| **options : dynamic-routing** | optional string, either **true** or **false** |
| **options : dynamic-routing-redistribute** | optional string |
| **options : dynamic-routing-redistribute-local-only** | optional string, either **true** or **false** |
| **options : dynamic-routing-vrf-name** | optional string |
| **options : dynamic-routing-vrf-id** | optional string, containing an integer, in range 1 to 4,294,967,295 |
| **options : dynamic-routing-no-learning** | optional string, either **true** or **false** |
| **options : dynamic-routing-v4-prefix-nexthop** | optional string |
| **options : dynamic-routing-v6-prefix-nexthop** | optional string |
| **options : ct-commit-all** | optional string, either **true** or **false** |
| **options : ic-route-filter-adv** | optional string |
| **options : ic-route-filter-learn** | optional string |
| **options : ic-route-deny-adv** | optional string |
| **options : ic-route-deny-learn** | optional string |
| **options : ic-route-ipv4-next-hop-ipv6** | optional string, either **true** or **false** |
| **options : disable_garp_rarp** | optional string, either **true** or **false** |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |
| *Transit router:* | |

**Details:**

**ports**: set of **Logical_Router_Port**s
The router's ports.

**static_routes**: set of **Logical_Router_Static_Route**s
Zero or more static routes for the router.

**policies**: set of **Logical_Router_Policy**s
Zero or more routing policies for the router.

**enabled**: optional boolean
This column is used to administratively set router state. If this column is empty or is set to **true**, the router is enabled. If this column is set to **false**, the router is disabled. A disabled router has all ingress and egress traffic dropped.

**nat**: set of **NAT**s
One or more NAT rules for the router. NAT rules only work on Gateway routers, and on distributed routers with one and only one distributed gateway port.

**load_balancer**: set of weak reference to **Load_Balancer**s
Set of load balancers associated to this logical router. Load balancer rules only work without limitations on the Gateway routers or routers with one and only one distributed gateway port (DGP). Load balancers will only work in scenarios that use more than one DGP when the multiple DGPs are associated with the same gateway chassis, this way this chassis can apply/maintain the conntrack state without problems. To use a load balancer in scenarios with DGPs associated with different gateway chassis (e.g. ECMP routes), consider using the **use_stateless_nat** option to **true** in the load balancer options column.

**load_balancer_group**: set of **Load_Balancer_Group**s
Set of load balancers groups associated to this logical router.

*Naming:*

These columns provide names for the logical router. From OVN's perspective, these names have no special meaning or purpose other than to provide convenience for human interaction with the northbound database. There is no requirement for the name to be unique. (For a unique identifier for a logical router, use its row UUID.)

(Originally, **name** was intended to serve the purpose of a human-friendly name, but the Neutron integration used it to uniquely identify its own router object, in the format **neutron−***uuid*. Later on, Neutron started propagating the friendly name of a router as **external_ids:neutron:router_name**. Perhaps this can be cleaned up someday.)

**name**: string
A name for the logical router.

**external_ids : neutron:router_name**: optional string
Another name for the logical router.

**copp**: optional weak reference to **Copp**
The control plane protection policy from table **Copp** used for metering packets sent to **ovn−controller** from logical ports of this router.

*Options:*

Additional options for the logical router.

**options : chassis**: optional string
If set, indicates that the logical router in question is a Gateway router (which is centralized) and resides in the set chassis. The same value is also used by **ovn−controller** to uniquely identify the chassis in the OVN deployment and comes from **external_ids:system−id** in the **Open_vSwitch** table of Open_vSwitch database.

The Gateway router can only be connected to a distributed router via a switch if SNAT and DNAT are to be configured in the Gateway router.

**options : dnat_force_snat_ip**: optional string
If set, indicates a set of IP addresses to use to force SNAT a packet that has already been DNATed in the gateway router. When multiple gateway routers are configured, a packet can potentially enter any of the gateway router, get DNATted and eventually reach the logical switch port. For the return traffic to go back to the same gateway router (for unDNATing), the packet needs a SNAT in the first place. This can be achieved by setting the above option with a gateway specific set of IP addresses. This option may have exactly one IPv4 and/or one IPv6 address on it, separated by a a space.

**options : lb_force_snat_ip**: optional string
If set, this option can take two possible type of values. Either a set of IP addresses or the string value - **router_ip**.

If a set of IP addresses are configured, it indicates to use to force SNAT a packet that has already been load-balanced in the gateway router. When multiple gateway routers are configured, a packet can potentially enter any of the gateway routers, get DNATted as part of the load-balancing and eventually reach the logical switch port. For the return traffic to go back to the same gateway router (for unDNATing), the packet needs a SNAT in the first place. This can be achieved by setting the above option with a gateway specific set of IP addresses. This option may have exactly one IPv4 and/or one IPv6 address on it, separated by a space character.

If it is configured with the value **router_ip**, then the load balanced packet is SNATed with the IP of router port (attached to the gateway router) selected as the destination after taking the routing decision.

**options : mcast_relay**: optional string, either **true** or **false**
Enables/disables IP multicast relay between logical switches connected to the logical router. Default: False.

**options : dynamic_neigh_routers**: optional string, either **true** or **false**
If set to **true**, the router will resolve neighbor routers' MAC addresses only by dynamic ARP/ND, instead of prepopulating static mappings for all neighbor routers in the ARP/ND Resolution stage. This reduces number of flows, but requires ARP/ND messages to resolve the IP-MAC bindings when needed. It is **false** by default. It is recommended to set to **true** when a large number of logical routers are connected to the same logical switch but most of them never need to send traffic between each other. By default, ovn-northd does not create mappings to NAT and load balancer addresess. However, for NAT and load balancer addresses that have the **add_route** option added, ovn-northd will create logical flows that map NAT and load balancer IP addresses to the appropriate MAC address. Setting *dynamic_neigh_routers* to **true** will prevent the automatic creation of these logical flows.

**options : always_learn_from_arp_request**: optional string, either **true** or **false**
This option controls the behavior when handling IPv4 ARP requests or IPv6 ND-NS packets - whether a dynamic neighbor (MAC binding) entry is added/updated.

**true** - Always learn the MAC-IP binding, and add/update the MAC binding entry.

**false** - If there is a MAC binding for that IP and the MAC is different, or, if TPA of ARP request belongs to any router port on this router, then update/add that MAC-IP binding. Otherwise, don't update/add entries.

It is **true** by default. It is recommended to set to **false** when a large number of logical routers are connected to the same logical switch but most of them never need to send traffic between each other, to reduce the size of the MAC binding table.

**options : requested-tnl-key**: optional string, containing an integer, in range 1 to 16,777,215
Configures the datapath tunnel key for the logical router. This is not needed because **ovn−northd** will assign an unique key for each datapath by itself. However, if it is configured, **ovn−northd** honors the configured value.

**options : snat-ct-zone**: optional string, containing an integer, in range 0 to 65,535
Use the requested conntrack zone for SNAT with this router. This can be useful if egress traffic from the host running OVN comes from both OVN and other sources. This way, OVN and the other sources can make use of the same conntrack zone.

**options : mac_binding_age_threshold**: optional string
Specifies the MAC binding aging thresholds based on CIDRs, with the format: *entry*[**;***entry*[...]], where each *entry* has the format: [*cidr***:**]*threshold*

- *cidr*: Can be either an IPv4 or IPv6 CIDR.
- *threshold*: Threshold value in seconds. MAC bindings with IP addresses matching the specified CIDR that exceed this timeout will be automatically removed.

If an *entry* is provided without an CIDR (just the threshold value), it specifies the default threshold for MAC bindings that don't match any of the given CIDRs. If there are multiple default threshold entries in the option, the behavior is undefined.

If there are multiple CIDRs matching a MAC binding IP, the one with the longest prefix length takes effect. If there are multiple entries with the same CIDR in the option, the behavior is undefined.

If no matching CIDR is found for a MAC binding IP, and no default threshold is specified, the behavior defaults to the original: the binding will not be removed based on age.

The value can also default to an empty string, which means that the aging threshold is disabled. Any string not in the above format is regarded as invalid and the aging is disabled.

Example: **192.168.0.0/16:300;192.168.10.0/24:0;fe80::/10:600;1200**

This sets a threshold of 300 seconds for MAC bindings with IP addresses in the 192.168.0.0/16 range, excluding the 192.168.1.0/24 range (for which the aging is disabled), a threshold of 600 seconds for MAC bindings with IP addresses in the fe80::/10 IPv6 range, and a default threshold of 1200 seconds for all other MAC bindings.

**options : ct-zone-limit**: optional string, containing an integer, in range 0 to 4,294,967,295
CT zone **limit** value for given **Logical_Router**. The value 0 means unlimited. When the option is not present the limit is not set and the zone limit is derived from OvS default datapath limit.

**options : dynamic-routing**: optional string, either **true** or **false**
If set to **true** then this **Logical_Router** can participate in dynamic routing with components outside of OVN. It will synchronize all routes to the soutbound **Advertised_Route** table that are relevant for the router. This includes:

- all "connected" routes implicitly created by networks associated with this Logical Router
- all **Logical_Router_Static_Route** that are applied to this Logical Router

Users will need to use the following settings to opt into individual route types that should be advertised. See:

- **options:dynamic-routing-redistribute** on Logical_Router
- **options:dynamic-routing-redistribute** on Logical_Router_Port

**options : dynamic-routing-redistribute**: optional string
Only relevant if **options:dynamic-routing** is set to **true**.

This is a list of elements separated by **,**.

If **connected** is in the list then northd will synchronize all "connected" routes to the southbound **Advertised_Route** table. "Connected" here means routes implicitly created by networks associated with the LRPs.

If **connected−as−host** is in the list then northd will enumerate all actively used individual IPs of a "connected" route and announce these IPs as host routes instead of announcing the subnet. This includes LSP and LRP addresses on the network as well as NAT entries of remove Logical_Routers on

this network. Setting this implies the setting **connected**. This setting can be used to:

- allow the fabric outside of OVN to drop traffic towards IP addresses that are not actually used. This traffic would otherwise hit this LR and then be dropped.
- If this LR has multiple LRPs connected to the fabric on different chassis: allows the fabric outside of OVN to steer packets to the chassis which already hosts this backing ip address.

If **static** is in the list then northd will synchronize all **Logical_Router_Static_Route** to the southbound **Advertised_Route** table.

If **lb** is in the list then northd will create entries in **Advertised_Route** table for each Load Balancer VIP on this router and it's neighboring routers. Neighboring routers are those that are either directly connected, via Logical Router Port, or those that are connected via shared Logical Switch.

If **nat** is in the list then northd will create entries in **Advertised_Route** table for each NAT's external IP on this router and it's neighboring routers. Neighboring routers are those that are either directly connected, via Logical Router Port, or those that are connected via shared Logical Switch.

If **hub−spoke** is in the list then northd will synchronize dynamic routes learned through OVN-IC from other routers into the **Advertised_Route** table, enabling hub-and-spoke propagation.

This value can be overwritten on a per LRP basis using **options:dynamic-routing-redistribute** on the Logical_Router_Port.

**options : dynamic-routing-redistribute-local-only**: optional string, either **true** or **false**
Only relevant if **options:dynamic-routing** is set to **true**.

This controls whether **ovn−controller** will advertise **Advertised_Route** records only on the chassis where their **tracked_port** is bound. Default: **false**.

**options : dynamic-routing-vrf-name**: optional string
Only relevant if **options:dynamic-routing** is set to **true**.

This defines the name of the vrf the ovn-controller will use to advertise and learn routes. If not set the vrf will be named "ovnvrf" with the **options:dynamic-routing-vrf-id** or datapath id of the Logical Router appended to it.

The vrf name must be a valid linux interface name. If it is too long the generated name will be used instead.

The vrf table id is not affected by this setting. For details see **options:dynamic-routing-maintain-vrf** on the Logical_Router.

**options : dynamic-routing-vrf-id**: optional string, containing an integer, in range 1 to 4,294,967,295
Only relevant if **options:dynamic-routing** is set to **true**.

This defines the vrf table id used by the specified vrf. When not set or set to invalid value datapath id of the Logical Router will be used instead.

This can also affect the vrf name. For details see **options:dynamic-routing-vrf-name** on the Logical_Router.

**options : dynamic-routing-no-learning**: optional string, either **true** or **false**
Only relevant if **options:dynamic-routing** is set to **true**.

This option disables route learning on a specific router and will also remove learned routes.

**options : dynamic-routing-v4-prefix-nexthop**: optional string
Only relevant if **options:dynamic-routing** is set to **true**.

This defines the nexthop used by IPv4 advertised routes instead of the routes being advertised as blackhole. This can be either a valid IPv4 or IPv6 address. When an IPv6 address is used, it enables RFC 5549-style routing where IPv4 traffic is forwarded over an IPv6-only peering link.

**options : dynamic-routing-v6-prefix-nexthop**: optional string
Only relevant if **options:dynamic-routing** is set to **true**.

This defines nexthop used by IPv6 advertised routes instead of the routes being advertised as blackhole. This can be only valid IPv6 address.

**options : ct-commit-all**: optional string, either **true** or **false**
When enabled the LR will commit traffic in a zone that is stateful. The traffic is not commited to both zones but it is selective based whether there is stateful DNAT/SNAT or both. The commit all will prevent issues with **ct.inv** packets as it will prevent the commit of reply traffic, which could happen in some cases. This also helps with HWOL as there shouldn't be any match on ct.new for established sessions as we will commit everything in addition to already existing stateful NATs and LBs.

**options : ic-route-filter-adv**: optional string
This option expects list of CIDRs delimited by "," that's present in the Logical Router. A route will be advertised if the route's prefix belongs to any of the CIDRs listed. This allows to filter CIDR prefixes in the process of advertising routes in **ovn−ic** daemon.

**options : ic-route-filter-learn**: optional string
This option expects list of CIDRs delimited by "," that's present in the Logical Router. A route will be learned if the route's prefix belongs to any of the CIDRs listed. This allows to filter CIDR prefixes in the process of learning routes in **ovn−ic** daemon.

**options : ic-route-deny-adv**: optional string
This option expects list of CIDRs delimited by "," that's present in the Logical Router. A route will not be advertised if the route's prefix belongs to any of the CIDRs listed. This allows to filter CIDR prefixes in the process of advertising routes in **ovn−ic** daemon.

**options : ic-route-deny-learn**: optional string
This option expects list of CIDRs delimited by "," that's present in the Logical Router. A route will not be learned if the route's prefix belongs to any of the CIDRs listed. This allows to filter CIDR prefixes in the process of learning routes in **ovn−ic** daemon.

**options : ic-route-ipv4-next-hop-ipv6**: optional string, either **true** or **false**
A boolean value that, when **true**, advertises this Logical Router's IPv4 routes into the global **OVN_IC_Southbound** database using the IPv6 address of the interconnect **Logical_Router_Port** as next hop instead of its IPv4 address ("IPv4 over IPv6"). This requires the interconnect port to have an IPv6 address; otherwise the IPv4 next hop is used as usual. The default value is **false**.

This option can be overridden per transit **Logical_Router_Port** by setting the option of the same name on that port.

**options : disable_garp_rarp**: optional string, either **true** or **false**
If set to **true**, GARP and RARP announcements are not sent by all the VIF peer ports of this logical router. The default value is **false**.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

*Transit router:*

In order to achieve status of **Transit Router** for **Logical_Router** there needs to be at least one **Logical_Router_Port** that is considered remote. The LRP can be **remote** only if it has **options:requested−chassis** set to chassis that is considered remote. See **Logical_Router_Port** for more details.

In order for the **Transit Router** to work properly all the tunnel keys for the **Transit Router** itself and its router ports need to match in all AZs e.g. TR in AZ1 and AZ2 need to have the same tunnel key. Remote ports for AZ2 in AZ1 need to have the same tunnel key as local ports in AZ2 and vice versa. It's also possible that the router has non-transit ports, without **options:requested−chassis** set; their keys also need to match in all

AZs.

The **Transit Router** behaves as distributed router which means that it has the same limitations for stateful flows like **NAT and LBs** and it will lose the CT state between AZs.

## QoS TABLE

Each row in this table represents one QoS rule for a logical switch that points to it through its **qos_rules** column. Two types of QoS are supported: DSCP marking and metering. A **match** with the highest-**priority** will have QoS applied to it. If the **action** column is specified, then matching packets will have DSCP marking applied. If the **bandwidth** column is specified, then matching packets will have metering applied. **action** and **bandwidth** are not exclusive, so both marking and metering by defined for the same QoS entry. If no row matches, packets will not have any QoS applied.

**Summary:**

| | |
|---|---|
| **priority** | integer, in range 0 to 32,767 |
| **direction** | string, either **from−lport** or **to−lport** |
| **match** | string |
| **action** | map of string-integer pairs, key either **dscp** or **mark**, value in range 0 to 4,294,967,295 |
| **bandwidth** | map of string-integer pairs, key either **burst** or **rate**, value in range 1 to 4,294,967,295 |
| **external_ids** | map of string-string pairs |

**Details:**

**priority**: integer, in range 0 to 32,767

The QoS rule's priority. Rules with numerically higher priority take precedence over those with lower. If two QoS rules with the same priority both match, then the one actually applied to a packet is undefined.

**direction**: string, either **from−lport** or **to−lport**

The value of this field is similar to **ACL** column in the OVN Northbound database's **ACL** table.

**match**: string

The packets that the QoS rules should match, in the same expression language used for the **match** column in the OVN Southbound database's **Logical_Flow** table. The **outport** logical port is only available in the **to−lport** direction (the **inport** is available in both directions).

**action**: map of string-integer pairs, key either **dscp** or **mark**, value in range 0 to 4,294,967,295

When **dscp** action is specified, matching flows will have have DSCP marking applied. When **mark** action is specified, matching flows will have packet marking applied.

- **dscp**: The value of this action should be in the range of 0 to 63 (inclusive).
- **mark**: The value of this action should be a positive integer.

**bandwidth**: map of string-integer pairs, key either **burst** or **rate**, value in range 1 to 4,294,967,295

When specified, matching packets will have bandwidth metering applied. Traffic over the limit will be dropped.

- **rate**: The value of rate limit in kbps.
- **burst**: The value of burst rate limit in kilobits. This is optional and needs to specify the **rate**.

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

## Mirror TABLE

Each row in this table represents a mirror that can be used for port mirroring. These mirrors are referenced by the **mirror_rules** column in the **Logical_Switch_Port** table.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **filter** | string, one of **both**, **from−lport**, or **to−lport** |
| **sink** | string |
| **type** | string, one of **erspan**, **gre**, **local**, or **lport** |
| **index** | integer |
| **mirror_rules** | set of **Mirror_Rule**s |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)

Represents the name of the mirror.

**filter**: string, one of **both**, **from−lport**, or **to−lport**

The value of this field represents selection criteria of the mirror. **to−lport** mirrors the packets coming into logical port. **from−lport** mirrors the packets going out of logical port. **both** mirrors for both directions.

**sink**: string

The value of this field represents the destination/sink of the mirror. If the *type* is **gre** or **erspan**, the value indicates the tunnel remote IP (either IPv4 or IPv6). For a *type* of **local**, this field defines a local interface on the OVS integration bridge to be used as the mirror destination. The interface must possess external-ids:mirror-id that matches this string.

**type**: string, one of **erspan**, **gre**, **local**, or **lport**

The value of this field specifies the mirror type - **gre**, **erspan**, **local** or **lport**.

**index**: integer

The value of this field represents the tunnel ID. If the configured tunnel type is **gre**, this field represents the **GRE** key value and if the configured tunnel type is **erspan** it represents the **erspan_idx** value. It is ignored if the type is **local**.

**mirror_rules**: set of **Mirror_Rule**s

The value of this field represents the mirror rule for filtering mirror traffic.

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### Mirror_Rule TABLE

Each row in this table represents a mirror rule that can be used for filtering of **lport** mirrored traffic.

**Summary:**

| | |
|---|---|
| **match** | string |
| **action** | string, either **mirror** or **skip** |
| **priority** | integer, in range 0 to 32,767 |

**Details:**

**match**: string

The match expression, describing which packets should be executed against Mirror Rule action. The Logical_Flow expression language is used.

**action**: string, either **mirror** or **skip**

The action to take when the Mirror Rule matches:

- **mirror**: Mirror the matched packet.
- **skip**: Do not mirror matched packet.

**priority**: integer, in range 0 to 32,767

The Mirror Rule priority. Rule with higher priority takes precedence over those with lower. A rule is uniquely identified by the priority and match string.

### Meter TABLE

Each row in this table represents a meter that can be used for QoS or rate-limiting.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **unit** | string, either **kbps** or **pktps** |
| **bands** | set of 1 or more **Meter_Band**s |
| **fair** | optional boolean |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)

A name for this meter.

Names that begin with "__" (two underscores) are reserved for OVN internal use and should not be added manually.

**unit**: string, either **kbps** or **pktps**

The unit for **rate** and **burst_rate** parameters in the **bands** entry. **kbps** specifies kilobits per second, and **pktps** specifies packets per second.

**bands**: set of 1 or more **Meter_Band**s

The bands associated with this meter. Each band specifies a rate above which the band is to take the action **action**. If multiple bands' rates are exceeded, then the band with the highest rate among the exceeded bands is selected.

**fair**: optional boolean

This column is used to further describe the desired behavior of the meter when there are multiple references to it. If this column is empty or is set to **false**, the rate will be shared across all rows that refer to the same Meter **name**. Conversely, when this column is set to **true**, each user of the same Meter will be rate-limited on its own.

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### Meter_Band TABLE

Each row in this table represents a meter band which specifies the rate above which the configured action should be applied. These bands are referenced by the **bands** column in the **Meter** table.

**Summary:**

| | |
|---|---|
| **action** | string, must be **drop** |
| **rate** | integer, in range 1 to 4,294,967,295 |
| **burst_size** | integer, in range 0 to 4,294,967,295 |
| **external_ids** | map of string-string pairs |

**Details:**

**action**: string, must be **drop**

The action to execute when this band matches. The only supported action is **drop**.

**rate**: integer, in range 1 to 4,294,967,295

The rate limit for this band, in kilobits per second or packets per second, depending on whether the parent **Meter** entry's **unit** column specified **kbps** or **pktps**.

**burst_size**: integer, in range 0 to 4,294,967,295

The maximum burst allowed for the band in kilobits or packets, depending on whether **kbps** or **pktps** was selected in the parent **Meter** entry's **unit** column. If the size is zero, the switch is free to select some reasonable value depending on its configuration.

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### Logical_Router_Port TABLE

A port within an L3 logical router.

Exactly one **Logical_Router** row must reference a given logical router port.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **networks** | set of strings |
| **mac** | string |
| **enabled** | optional boolean |
| **dhcp_relay** | optional **DHCP_Relay** |
| **options : dhcpv4_primary_ip** | optional string |
| *Distributed Gateway Ports:* | |
| **ha_chassis_group** | optional **HA_Chassis_Group** |
| **gateway_chassis** | set of **Gateway_Chassis**es |
| *Options for Physical VLAN MTU Issues:* | |
| **options : reside-on-redirect-chassis** | optional string, either **true** or **false** |
| **options : redirect-type** | optional string, either **bridged** or **overlay** |
| **ipv6_prefix** | set of strings |
| *ipv6_ra_configs:* | |
| **ipv6_ra_configs : address_mode** | optional string |
| **ipv6_ra_configs : router_preference** | optional string |
| **ipv6_ra_configs : route_info** | optional string |
| **ipv6_ra_configs : mtu** | optional string |
| **ipv6_ra_configs : send_periodic** | optional string |
| **ipv6_ra_configs : max_interval** | optional string |
| **ipv6_ra_configs : min_interval** | optional string |
| **ipv6_ra_configs : rdnss** | optional string |
| **ipv6_ra_configs : dnssl** | optional string |
| *Options:* | |
| **options : mcast_flood** | optional string, either **true** or **false** |
| **options : requested-tnl-key** | optional string, containing an integer, in range 1 to 32,767 |
| **options : prefix_delegation** | optional string, either **true** or **false** |
| **options : prefix** | optional string, either **true** or **false** |
| **options : route_table** | optional string |
| **options : gateway_mtu** | optional string, containing an integer, in range 68 to 65,535 |
| **options : routing-protocol-redirect** | optional string |
| **options : routing-protocols** | optional string |
| **options : gateway_mtu_bypass** | optional string |
| **options : ic-route-tag** | optional string |
| **options : ic-route-filter-tag** | optional string |
| **options : ic-route-learn-tag-rules** | optional string |
| **options : requested-chassis** | optional string |
| **options : requested-encap-ip** | optional string |
| **options : dynamic-routing-redistribute** | optional string |
| **options : dynamic-routing-advertise** | optional string |
| **options : dynamic-routing-redistribute-local-only** | optional string, either **true** or **false** |
| **options : dynamic-routing-maintain-vrf** | optional string, either **true** or **false** |
| **options : dynamic-routing-port-name** | optional string |

| | |
|---|---|
| **options : ic-route-filter-adv** | optional string |
| **options : ic-route-filter-learn** | optional string |
| **options : ic-route-deny-adv** | optional string |
| **options : ic-route-deny-learn** | optional string |
| **options : ic-route-ipv4-next-hop-ipv6** | optional string, either **true** or **false** |
| **options : dynamic-routing-no-learning** | optional string, either **true** or **false** |
| *Attachment:* | |
| **peer** | optional string |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |
| *Status:* | |
| **status : hosting-chassis** | optional string |

**Details:**

**name**: string (must be unique within table)

A name for the logical router port.

In addition to provide convenience for human interaction with the northbound database, this column is used as reference by its patch port in **Logical_Switch_Port** or another logical router port in **Logical_Router_Port**.

A logical router port may not have the same name as a logical switch port, but the database schema cannot enforce this.

**networks**: set of strings

The IP addresses and netmasks of the router. For example, **192.168.0.1/24** indicates that the router's IP address is 192.168.0.1 and that packets destined to 192.168.0.*x* should be routed to this port. These are optional.

A logical router port always adds a link-local IPv6 address (fe80::/64) automatically generated from the interface's MAC address using the modified EUI−64 format.

**mac**: string

The Ethernet address that belongs to this router port.

**enabled**: optional boolean

This column is used to administratively set port state. If this column is empty or is set to **true**, the port is enabled. If this column is set to **false**, the port is disabled. A disabled port has all ingress and egress traffic dropped.

**dhcp_relay**: optional **DHCP_Relay**

This column is used to enabled DHCP Relay. Please refer to **DHCP_Relay** table.

**options : dhcpv4_primary_ip**: optional string

Optional. Selects which of the logical router port's IPv4 addresses **ovn−northd** uses as the relay agent address (**giaddr**) in DHCP packets relayed via this port (see **dhcp_relay**). The value must be a bare IPv4 host address (no prefix length, e.g. **10.0.0.1**) that equals one of the addresses configured in **networks**. If the value is not a valid IPv4 address or does not match one of the addresses in **networks**, **ovn−northd** skips emitting DHCP relay flows for this port.

When unset, **ovn−northd** defaults to the first address in **networks** as ordered by OVSDB: the CIDR that sorts first as a string. This default may not match the address intended as the DHCP relay address of the port, so on multi-IP ports this option should be set explicitly. On single-IP ports the only configured address is always used and the option may be left unset.

*Distributed Gateway Ports:*

Gateways, as documented under **Gateways** in the OVN architecture guide, provide limited connectivity between logical networks and physical ones. OVN support multiple kinds of gateways. The **Logical_Router_Port** table can be used two different ways to configure *distributed gateway ports*, which are one

kind of gateway. These two forms of configuration exist for historical reasons. Both of them produce the same kind of OVN southbound records and the same behavior in practice.

If either of these are set, this logical router port represents a distributed gateway port that connects this router to a logical switch with a **localnet** port or a connection to another OVN deployment.

Also mentioned in the OVN architecture guide, distributed gateway ports can also be used for scalability reasons in deployments where logical switches are dedicated to chassises rather than distributed.

The preferred way to configure a gateway is **ha_chassis_group**, but **gateway_chassis** is also supported for backward compatibility. Only one of these should be set at a time on a given LRP, since they configure the same features.

Even when a gateway is configured, the logical router port still effectively resides on each chassis. However, due to the implications of the use of L2 learning in the physical network, as well as the need to support advanced features such as one-to-many NAT (aka IP masquerading), a subset of the logical router processing is handled in a centralized manner on the gateway chassis.

There can be more than one distributed gateway ports configured on each logical router, each connecting to different L2 segments. Load-balancing is not yet supported on logical routers with more than one distributed gateway ports.

For each distributed gateway port, it may have more than one gateway chassises. When more than one gateway chassis is specified, OVN only uses one at a time. OVN can rely on OVS BFD implementation to monitor gateway connectivity, preferring the highest-priority gateway that is online. Priorities are specified in the **priority** column of **Gateway_Chassis** or **HA_Chassis**.

**ovn−northd** programs the **external_mac** rules specified in the LRP's LR into the peer logical switch's destination lookup on the chassis where the **logical_port** resides. In addition, the logical router's MAC address is automatically programmed in the peer logical switch's destination lookup flow on the gateway chasssis. If it is desired to generate gratuitous ARPs for NAT addresses, then set the peer LSP's **options:nat-addresses** to **router**.

OVN 20.03 and earlier supported a third way to configure distributed gateway ports using **options:redirect−chassis** to specify the gateway chassis. This method is no longer supported. Any remaining users should switch to one of the newer methods instead. A **gateway_chassis** may be easily configured from the command line, e.g. **ovn−nbctl lrp−set−gateway−chassis** *lrp chassis*.

**ha_chassis_group**: optional **HA_Chassis_Group**
Designates an **HA_Chassis_Group** to provide gateway high availability.

**gateway_chassis**: set of **Gateway_Chassis**es
Designates one or more **Gateway_Chassis** for the logical router port.

*Options for Physical VLAN MTU Issues:*

MTU issues arise in mixing tunnels with logical networks that are bridged to a physical VLAN. For an explanation of the MTU issues, see **Physical VLAN MTU Issues** in the OVN architecture document. The following options, which are alternatives, provide solutions. Both of them cause packets to be sent over **localnet** instead of tunnels, but they differ in whether some or all packets are sent this way. The most prominent tradeoff between these options is that **reside−on−redirect−chassis** is easier to configure and that **redirect−type** performs better for east-west traffic.

**options : reside-on-redirect-chassis**: optional string, either **true** or **false**
If set to **true**, this option forces all traffic across the logical router port to pass through the gateway chassis using a hop across a **localnet** port. This changes behavior in two ways:

- Without this option, east-west traffic passes directly between source and destination chassis (or even within a single chassis, for co-located VMs). With this option, all east-west traffic passes through the gateway chassis.

- Without this option, traffic between the gateway chassis and other chassis is encapsulated in tunnels. With this option, traffic passes over a **localnet** interface.

This option may usefully be set only on logical router ports that connect a distributed logical router to a logical switch with VIFs. It should not be set on a distributed gateway port.

OVN honors this option only if the logical router has one and only one distributed gateway port and if the LRP's peer switch has a **localnet** port.

**options : redirect-type**: optional string, either **bridged** or **overlay**
If set to **bridged** on a distributed gateway port, this option causes OVN to redirect packets to the gateway chassis over a **localnet** port instead of a tunnel. The relevant chassis must share a **localnet** port.

This feature requires the administrator or the CMS to configure each participating chassis with a unique Ethernet address for the logical router by setting **ovn−chassis−mac−mappings** in the Open vSwitch database, for use by **ovn−controller**.

Setting this option to **overlay** or leaving it unset has no effect. This option may usefully be set only on a distributed gateway port when there is one and only one distributed gateway port on the logical router. It is otherwise ignored.

**ipv6_prefix**: set of strings
This column contains IPv6 prefix obtained by prefix delegation router according to RFC 3633

*ipv6_ra_configs:*

This column defines the IPv6 ND RA address mode and ND MTU Option to be included by **ovn−controller** when it replies to the IPv6 Router solicitation requests.

**ipv6_ra_configs : address_mode**: optional string
The address mode to be used for IPv6 address configuration. The supported values are:

- **slaac**: Address configuration using Router Advertisement (RA) packet. The IPv6 prefixes defined in the **Logical_Router_Port** table's **networks** column will be included in the RA's ICMPv6 option - Prefix information.
- **dhcpv6_stateful**: Address configuration using DHCPv6.
- **dhcpv6_stateless**: Address configuration using Router Advertisement (RA) packet. Other IPv6 options are provided by DHCPv6.

**ipv6_ra_configs : router_preference**: optional string
Default Router Preference (PRF) indicates whether to prefer this router over other default routers (RFC 4191). Possible values are:

- HIGH: mapped to 0x01 in RA PRF field
- MEDIUM: mapped to 0x00 in RA PRF field
- LOW: mapped to 0x11 in RA PRF field

**ipv6_ra_configs : route_info**: optional string
Route Info is used to configure Route Info Option sent in Router Advertisement according to RFC 4191. Route Info is a comma separated string where each field provides PRF and prefix for a given route (e.g: HIGH-aef1::11/48,LOW-aef2::11/96) Possible PRF values are:

- HIGH: mapped to 0x01 in RA PRF field
- MEDIUM: mapped to 0x00 in RA PRF field
- LOW: mapped to 0x11 in RA PRF field

**ipv6_ra_configs : mtu**: optional string
The recommended MTU for the link. Default is 0, which means no MTU Option will be included in RA packet replied by ovn-controller. Per RFC 2460, the mtu value is recommended no less than

1280, so any mtu value less than 1280 will be considered as no MTU Option.

**ipv6_ra_configs : send_periodic**: optional string
If set to true, then this router interface will send router advertisements periodically. This option has no effect if **ipv6_ra_configs:address_mode** is not set. The default is false.

**ipv6_ra_configs : max_interval**: optional string
The maximum number of seconds to wait between sending periodic router advertisements. This option has no effect if **ipv6_ra_configs:send_periodic** is false. The default is 600.

**ipv6_ra_configs : min_interval**: optional string
The minimum number of seconds to wait between sending periodic router advertisements. This option has no effect if **ipv6_ra_configs:send_periodic** is false. The default is one-third of **ipv6_ra_configs:max_interval**, i.e. 200 seconds if that key is unset.

**ipv6_ra_configs : rdnss**: optional string
IPv6 address of RDNSS server announced in RA packets. At the moment OVN supports just one RDNSS server.

**ipv6_ra_configs : dnssl**: optional string
DNS Search List announced in RA packets. Multiple DNS Search List must be 'comma' separated (e.g. "a.b.c, d.e.f")

*Options:*

Additional options for the logical router port.

**options : mcast_flood**: optional string, either **true** or **false**
If set to **true**, multicast traffic (including reports) are unconditionally forwarded to the specific port.

This option applies when the port is part of a logical router which has **options**:mcast_relay set to **true**.

Default: **false**.

**options : requested-tnl-key**: optional string, containing an integer, in range 1 to 32,767
Configures the port binding tunnel key for the port. Usually this is not needed because **ovn−northd** will assign an unique key for each port by itself. However, if it is configured, **ovn−northd** honors the configured value.

**options : prefix_delegation**: optional string, either **true** or **false**
If set to **true**, enable IPv6 prefix delegation state machine on this logical router port (RFC3633). IPv6 prefix delegation is available just on a gateway router or on a gateway router port.

**options : prefix**: optional string, either **true** or **false**
If set to **true**, this interface will receive an IPv6 prefix according to RFC3663

**options : route_table**: optional string
Designates lookup for **Logical_Router_Static_Routes** with the specified **route_table** value. See detailed explanation for routes lookup behavior in **Logical_Router_Static_Route:route_table** field description.

**options : gateway_mtu**: optional string, containing an integer, in range 68 to 65,535
If set, logical flows will be added to router pipeline to check packet length. If packet length is greater than the value set, ICMPv4 type 3 (Destination Unreachable) code 4 (Fragmentation Needed and Don't Fragment was Set) or ICMPv6 type 2 (Packet Too Big) code 0 (no route to destination) packets will be generated. This allows for Path MTU Discovery.

**options : routing-protocol-redirect**: optional string
This option expects a name of a Logical Switch Port that's present in the peer's Logical Switch. If set, it causes any traffic that's destined for Logical Router Port's IP addresses (including its IPv6 LLA) and the ports associated with routing protocols defined ip **routing−protocols** option, to be redirected to the specified Logical Switch Port. This allows external routing daemons to be bound to

a port in OVN's Logical Switch and act as if they were listening on Logical Router Port's IP addresses.

When used with dynamic routing (when **options:dynamic-routing** is set to **true**), if the specified Logical Switch Port is bound locally and connected to a veth pair, **ovn−controller** is able to automatically discover the peer interface name and use it for route learning. This removes the need to manually configure **options:dynamic-routing-port-name** and/or **external_ids:dynamic-routing-port-mapping** for veth-based routing daemon integrations. If those options are set they always take precedence over auto-discovery.

The auto-discovery feature relies on the peer interface index being reported in the **Open_vSwitch Interface** table **status:peer_ifindex** key, which is only populated for veth devices on Linux systems by a sufficiently recent version of Open vSwitch. If the bound interface is not a veth device, if the peer cannot be resolved, or if OVS does not report **status:peer_ifindex**, the system will fallback to learning routes from all interfaces on the Logical Router Port.

**options : routing-protocols**: optional string

This option expects a comma-separated list of routing, and routing-related protocols, whose control plane traffic will be redirected to a port specified in **routing−protocol−redirect** option. Currently supported options are:

- **BGP** (forwards TCP port 179)
- **BFD** (forwards UDP port 3784)

Note that for BGP to work in "unnumbered mode" (advertising IPv4 routes over IPv6 network, with automatic on-link peer discovery), Logical Router Port needs to enable sending of periodic IPv6 Router Announcements (see the **ipv6_ra_configs:send_periodic**). Recommended minimal configuration of periodic RAs for BGP unnumbered:

- **ipv6_ra_configs:address_mode = slaac** (Any valid value is OK, but the option needs to be set)
- **ipv6_ra_configs:send_periodic = true**
- **ipv6_ra_configs:max_interval = 10**
- **ipv6_ra_configs:min_interval = 5**

Feel free to adjust max and min interval values, but be aware that they influence speed at which the initial BGP session is established. With the above values, the session will be established in 5 to 10 seconds. Please refer to RFC 8950 for more details about advertising IPv4 networks over IPv6 next-hop addresses.

**options : gateway_mtu_bypass**: optional string

When configured, represents a match expression, in the same expression language used for the **match** column in the OVN Southbound database's **Logical_Flow** table. Packets matching this expression will bypass the length check configured through the **options:gateway_mtu** option.

**options : ic-route-tag**: optional string

This option expects a name of a route-tag that's present in the Logical Router Port. If set, it causes any route advertised by the Logical Router Port to include the **route−tag** in the **external_ids** register of the advertised route entry in the **Route** table of the **OVN_IC_Southbound** database. This allows to tag and filter route tags in the process of advertising and learning routes in **ovn−ic** daemon.

**options : ic-route-filter-tag**: optional string

This option expects a name of a filtered route-tag that's present in the Logical Router Port. If set, it causes any route learned by the Logical Router Port with the **route−tag** present in the external_ids register of the advertised route entry in the **Route** table of the **OVN_IC_Southbound** database, will be filtered and not learned by the **ovn−ic** daemon.

This option is deprecated and will be removed in the 28.09 release. Use **options:ic-route-learn-tag-rules** instead, which supports more than one route tag and which, when set, causes this option to be

ignored.

**options : ic-route-learn-tag-rules**: optional string
This option controls, based on route tags, which routes the **ovn−ic** daemon learns through this Logical Router Port. It expects a value in one of the following two forms, where **tags** is a comma-separated list of route-tag names, for example **allow:vpc1,vpc2**:

- **allow:tags** - only routes whose **ic−route−tag** matches one of the listed tags are learned. Every other route, including routes that carry no **ic−route−tag** at all, is not learned.
- **block:tags** - routes whose **ic−route−tag** matches one of the listed tags are not learned. Every other route, including routes that carry no **ic−route−tag** at all, is learned.

The **ic−route−tag** of a route is the one present in the **external_ids** register of the advertised route entry in the **Route** table of the **OVN_IC_Southbound** database. As commas separate the tags, a route tag cannot contain a comma. Values that use neither the **allow:** nor the **block:** prefix, as well as values with an empty tag list, are invalid and are ignored (no route is filtered by this option).

The two forms are mutually exclusive: only the leading prefix is honored and everything after it is a route tag name. In **allow:vpc1,block:vpc2**, for example, the allowed tags are **vpc1** and **block:vpc2**.

This option supersedes the deprecated **options:ic-route-filter-tag** option: whenever a valid value is set here, the route tags to learn are the ones defined by this option only and **options:ic-route-filter-tag** is ignored.

**options : requested-chassis**: optional string
If set, identifies a specific chassis (by name or hostname) that is allowed to bind this port. This option is valid only for chassis that have **options:is−remote=true**, in other words for chassis that are in different Availability zone. The option accepts only single value.

By assigning remote chassis the **Logical_Router** gains status of **Transit Router** see **Logical_Router** table for more details.

**options : requested-encap-ip**: optional string
Requests the encapsulation IP address for the port binding. If set, **ovn−northd** uses this IP to select the **Encap** entry written to **encap**. This is intended for ports without a local OVS interface, e.g. remote transit router ports in ovn-kubernetes interconnect mode.

**options : dynamic-routing-redistribute**: optional string
Only relevant if **options:dynamic-routing** on the respective Logical_Router is set to **true**.

This is a list of elements separated by **,**.

If **connected** is in the list then northd will synchronize all "connected" routes to the southbound **Advertised_Route** table. "Connected" here means routes implicitly created by networks associated with the LRPs.

If **connected−as−host** is in the list then northd will enumerate all actively used individual IPs of a "connected" route and announce these IPs as host routes instead of announcing the subnet. This includes LSP and LRP addresses on the network as well as NAT entries of remove Logical_Routers on this network. Setting this implies the setting **connected**. This setting can be used to:

- allow the fabric outside of OVN to drop traffic towards IP addresses that are not actually used. This traffic would otherwise hit this LR and then be dropped.
- If this LR has multiple LRPs connected to the fabric on different chassis: allows the fabric outside of OVN to steer packets to the chassis which already hosts this backing ip address.

If **static** is in the list then northd will synchronize all **Logical_Router_Static_Route** to the southbound **Advertised_Route** table.

If **lb** is in the list then northd will create entries in **Advertised_Route** table for each Load Balancer VIP on this port's router, and it's neighboring routers. Neighboring routers are those that are either

directly connected to this Logical Router Port, or those that are connected via shared Logical Switch.

If **nat** is in the list then northd will create entries in **Advertised_Route** table for each NAT's external IP on this port's router, and it's neighboring routers. Neighboring routers are those that are either directly connected to this Logical Router Port, or those that are connected via shared Logical Switch.

If **hub−spoke** is in the list then northd will synchronize dynamic routes learned through OVN-IC from other routers into the **Advertised_Route** table, enabling hub-and-spoke propagation.

If not set the value from **options:dynamic-routing-redistribute** on the Logical_Router will be used.

**options : dynamic-routing-advertise**: optional string
If the CMS sets **options:dynamic-routing-redistribute** to **connected−as−host**, this parameter is used by the CMS to enable/disable the advertisement of the current host IP via the **Advertised_Route** table. Default: **true**.

**options : dynamic-routing-redistribute-local-only**: optional string, either **true** or **false**
Only relevant if **options:dynamic-routing** is set to **true**.

This controls whether **ovn−controller** will advertise **Advertised_Route** records only on the chassis where their **tracked_port** is bound.

If not set the value from **options:dynamic-routing-redistribute-local-only** on the **Logical_Router** will be used.

**options : dynamic-routing-maintain-vrf**: optional string, either **true** or **false**
Only relevant if **options:dynamic-routing** on the respective Logical_Router is set to **true**.

If this LRP is bound to a specific chassis then the ovn-controller of this chassis will maintain a vrf. This vrf will contain all the routes that should be announced from this LRP. Unless **options:dynamic-routing-vrf-name** is set the vrf will be named "ovnvrf" with the **options:dynamic-routing-vrf-id** or datapath id of the Logical Router appended to it.

If the setting is not set or false the ovn-controller will expect this VRF to already exist. Some tooling outside of OVN needs to ensure this.

The VRF table ID is the same as the tunnel key of the Logical_Router datapath. If this setting is false the tooling outside of OVN needs to ensure that this is the case.

**options : dynamic-routing-port-name**: optional string
Only relevant if **options:dynamic-routing** on the respective Logical_Router is set to **true**. Only learn routes associated with the interface locally bound to the LSP or LRP specified here. This allows a single chassis to learn different routes on separate LRPs bound to this chassis. This is usefully e.g. in the case of a chassis with multiple links towards the network fabric where all of them run BGP individually. This option allows to have a 1:1 mapping between a single LRP and an individual link. If the port referenced by this name is bound locally on the ovn-controller we lookup the linux interface name of this port. This interface name is then used for the route filtering, so only routes that have this interface as nexthop will be learned. As there might not always be such a port bound on the ovn-controller this value can also be an arbitrary string. The ovn-controller will lookup the port name in the local **external_ids:dynamic-routing-port-mapping**. This is a list separated by **,** that contains **port=interfacename** pairs. If a match is found in there the configured interface name is used instead of the autodiscovery. Also it is then irrelevant if the port is bound locally.

**options : ic-route-filter-adv**: optional string
This option expects list of CIDRs delimited by "," that's present in the Logical Router Port. A route will be advertised if the route's prefix belongs to any of the CIDRs listed. This allows to filter CIDR prefixes in the process of advertising routes in **ovn−ic** daemon.

**options : ic-route-filter-learn**: optional string
This option expects list of CIDRs delimited by "," that's present in the Logical Router Port. A route will be learned if the route's prefix belongs to any of the CIDRs listed. This allows to filter CIDR

prefixes in the process of learning routes in **ovn−ic** daemon.

**options : ic-route-deny-adv**: optional string
This option expects list of CIDRs delimited by "," that's present in the Logical Router Port. A route will not be advertised if the route's prefix belongs to any of the CIDRs listed. This allows to filter CIDR prefixes in the process of advertising routes in **ovn−ic** daemon.

**options : ic-route-deny-learn**: optional string
This option expects list of CIDRs delimited by "," that's present in the Logical Router Port. A route will not be learned if the route's prefix belongs to any of the CIDRs listed. This allows to filter CIDR prefixes in the process of learning routes in **ovn−ic** daemon.

**options : ic-route-ipv4-next-hop-ipv6**: optional string, either **true** or **false**
Overrides the **ic−route−ipv4−next−hop−ipv6** option of the **Logical_Router** for routes advertised through this transit port. When **true**, IPv4 routes are advertised using this port's IPv6 address as next hop ("IPv4 over IPv6"); when **false**, the IPv4 next hop is used. If unset, the Logical Router option applies.

**options : dynamic-routing-no-learning**: optional string, either **true** or **false**
Only relevant if **options:dynamic-routing** is set to **true**.

This option disables route learning on a specific router port and will also remove learned routes. It also has priority over the router version of this option.

*Attachment:*

A given router port serves one of two purposes:

- To attach a logical switch to a logical router. A logical router port of this type is referenced by exactly one **Logical_Switch_Port** of type **router**. The value of **name** is set as **router−port** in column **options** of **Logical_Switch_Port**. In this case **peer** column is empty.
- To connect one logical router to another. This requires a pair of logical router ports, each connected to a different router. Each router port in the pair specifies the other in its **peer** column. No **Logical_Switch** refers to the router port.

**peer**: optional string
For a router port used to connect two logical routers, this identifies the other router port in the pair by **name**.

For a router port attached to a logical switch, this column is empty.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

The **ovn−northd** program copies all these pairs into the **external_ids** column of the **Port_Binding** table in **OVN_Southbound** database.

*Status:*

Additional status about the logical router port.

**status : hosting-chassis**: optional string
This option is populated by **ovn−northd**.

When a distributed gateway port is bound to a location in the OVN Southbound database **Port_Binding ovn−northd** will populate this key with the name of the Chassis that is currently hosting this port.

### Logical_Router_Static_Route TABLE

Each record represents a static route.

When multiple routes match a packet, the longest-prefix match is chosen. For a given prefix length, a **dst−ip** route is preferred over a **src−ip** route.

When there are ECMP routes, i.e. multiple routes with same prefix and policy, one of them will be selected based on the 5-tuple hashing of the packet header.

**Summary:**

| | |
|---|---|
| **ip_prefix** | string |
| **policy** | optional string, either **dst−ip** or **src−ip** |
| **nexthop** | string |
| **output_port** | optional string |
| **bfd** | optional weak reference to **BFD** |
| **selection_fields** | set of strings, one of **eth_dst**, **eth_src**, **ip_dst**, **ip_proto**, **ip_src**, **ipv6_dst**, **ipv6_src**, **tp_dst**, or **tp_src** |
| **route_table** | string |
| **external_ids : ic-learned-route** | optional string |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |
| *Common options:* | |
| **options** | map of string-string pairs |
| **options : ecmp_symmetric_reply** | optional string |
| **options : origin** | optional string |
| **options : dynamic-routing-advertise** | optional string |

**Details:**

**ip_prefix**: string

IP prefix of this route (e.g. 192.168.100.0/24).

**policy**: optional string, either **dst−ip** or **src−ip**

If it is specified, this setting describes the policy used to make routing decisions. This setting must be one of the following strings:

- **src−ip**: This policy sends the packet to the **nexthop** when the packet's source IP address matches **ip_prefix**.
- **dst−ip**: This policy sends the packet to the **nexthop** when the packet's destination IP address matches **ip_prefix**.

If not specified, the default is **dst−ip**.

**nexthop**: string

Nexthop IP address for this route. Nexthop IP address should be the IP address of a connected router port or the IP address of a logical port or can be set to **discard** for dropping packets which match the given route.

**output_port**: optional string

The name of the **Logical_Router_Port** via which the packet needs to be sent out. This is optional and when not specified, OVN will automatically figure this out based on the **nexthop**. When this is specified and there are multiple IP addresses on the router port and none of them are in the same subnet of **nexthop**, OVN chooses the first IP address as the one via which the **nexthop** is reachable.

**bfd**: optional weak reference to **BFD**

Reference to **BFD** row if the route has associated a BFD session

**selection_fields**: set of strings, one of **eth_dst**, **eth_src**, **ip_dst**, **ip_proto**, **ip_src**, **ipv6_dst**, **ipv6_src**, **tp_dst**, or **tp_src**

ECMP routes use OpenFlow groups of type **select** to pick a nexthop among the list of available

nexthops. OVS supports two selection methods: **dp_hash** and **hash** for hash computation and selecting the buckets of a group. OVN by default uses **dp_hash**. In order to use the **hash** selection method, specify the allowed match fields in selection fields. Please see the OVS documentation (man ovs-ofctl) for more details on selection methods and fields.

To match on Layer 4 ports use **tp_src** and **tp_dst**. This match is applicable only for TCP, UDP, SCTP and will be ignored for all other IP packets. When matching on Layer 4 ports, match on ip_proto will be implicitly added in the select action.

Example: **{ip_proto,ip_src,ip_dst}** for a 3-tuple match. Example: **{ip_proto,ipv6_src,ipv6_dst}** for an IPv6 match. Example: **{ip_proto,ip_src,ip_dst,tp_src,tp_dst}**. Example: **{ip_src,ip_dst,ipv6_src,ipv6_dst,tp_src,tp_dst}**.

**route_table**: string
Specify any string to assign a route to a separate routing table. When a Logical Router Port has a configured value in **options:route_table**, only static routes with the same routing table value are considered. A more detailed description of the route lookup behavior is provided below:

When a packet enters Logical Router (**LR**), it examines the following list of routes:

- All routes to directly connected networks of the **LR** (including networks that are learned from other availability zones within the same **LR** through OVN-IC).
- All static routes of the **LR** that have the same **route_table** field value as that of the Logical Router Port's **options:route_table** (If the option is absent, static routes with an empty **route_table** field are considered).

From the resulting list of routes, the route with the longest prefix match takes precedence over others.

**external_ids : ic-learned-route**: optional string
**ovn−ic** populates this key if the route is learned from the global **OVN_IC_Southbound** database. In this case the value will be set to the uuid of the row in **Route** table of the **OVN_IC_Southbound** database.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

*Common options:*

**options**: map of string-string pairs
This column provides general key/value settings. The supported options are described individually below.

**options : ecmp_symmetric_reply**: optional string
If true, then

- New ingress-originated traffic that arrives over this route will have its reply traffic bypass ECMP route selection and will be sent out this route instead.
- For the egress-originated traffic, the ingress reply traffic route gets saved. And the subsequent traffic will bypass ECMP route selection and instead be sent out the same route.

Note that this option overrides any rules set in the **Logical_Router_policy** table. This option only works on gateway routers (routers that have **options:chassis** set).

**options : origin**: optional string
In case ovn-interconnection has been learned this route, it will have its origin set: either "connected" or "static". This key is supposed to be written only by **ovn−ic** daemon. ovn-northd then checks this value when generating Logical Flows. **Logical_Router_Static_Route** records with same **ip_prefix** within same Logical Router will have next lookup order based on **origin** key value:

1. connected
2. static

**options : dynamic-routing-advertise**: optional string
If the CMS sets **options:dynamic-routing-redistribute** to **static**, this parameter can be used to enable/disable the advertisement of the current static route via the **Advertised_Route** table. Default: **true**.

### Logical_Router_Policy TABLE

Each row in this table represents one routing policy for a logical router that points to it through its **policies** column. The **action** column for the highest-**priority** matching row in this table determines a packet's treatment. If no row matches, packets are allowed by default. (Default-deny treatment is possible: add a rule with **priority** 0, **1** as **match**, and **drop** as **action**.)

**Summary:**

| | |
|---|---|
| **priority** | integer, in range 0 to 32,767 |
| **chain** | optional string |
| **match** | string |
| **action** | string, one of **allow**, **drop**, **jump**, or **reroute** |
| **jump_chain** | optional string |
| **nexthop** | optional string |
| **nexthops** | set of strings |
| **output_port** | optional weak reference to **Logical_Router_Port** |
| **bfd_sessions** | set of weak reference to **BFD**s |
| **options : pkt_mark** | optional string |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**priority**: integer, in range 0 to 32,767

The routing policy's priority. Rules with numerically higher priority take precedence over those with lower. A rule is uniquely identified by the priority, chain and match string.

**chain**: optional string

The routing policy rule's chain name. Only rules with empty chain name are traversed by default. Other chains are traversed as response to jump action.

**match**: string

The packets that the routing policy should match, in the same expression language used for the **match** column in the OVN Southbound database's **Logical_Flow** table.

By default all traffic is allowed. When writing a more restrictive policy, it is important to remember to allow flows such as ARP and IPv6 neighbor discovery packets.

**action**: string, one of **allow**, **drop**, **jump**, or **reroute**

The action to take when the routing policy matches:

- **allow**: Forward the packet.
- **drop**: Silently drop the packet.
- **reroute**: Reroute packet to **nexthop** or **nexthops**.
- **jump**: Start examining rules that have the same **chain** value as specified in **jump_chain**.

**jump_chain**: optional string

The routing policy rule's chain name selected to be examined next.

**nexthop**: optional string

Note: This column is deprecated in favor of **nexthops** and is ignored by northd.

Next-hop IP address for this route, which should be the IP address of a connected router port or the IP address of a logical port.

**nexthops**: set of strings

Next-hop ECMP IP addresses for this route. Each IP in the list should be the IP address of a connected router port or the IP address of a logical port.

One IP from the list is selected as next hop.

**output_port**: optional weak reference to **Logical_Router_Port**

The **Logical_Router_Port** via which the packet needs to be sent out. This is optional and when not specified, OVN will automatically figure this out based on the **nexthops** column. When this is specified and there are multiple IP addresses on the router port and none of them are in the same subnet of **nexthops**, OVN chooses the first IP address as the one via which the **nexthops** are reachable. NOTE: for now OVN does not support configuring the output port on ECMP reroute policies (with more than one value in the **nexthops** column).

**bfd_sessions**: set of weak reference to **BFD**s

Reference to **BFD** row if the route policy has associated some BFD sessions.

**options : pkt_mark**: optional string

Marks the packet with the value specified when the router policy is applied. CMS can inspect this packet marker and take some decisions if desired. This value is not preserved when the packet goes out on the wire.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

## NAT TABLE

Each record represents a NAT rule.

**Summary:**

| | |
|---|---|
| **type** | string, one of **dnat**, **dnat_and_snat**, or **snat** |
| **external_ip** | string |
| **external_mac** | optional string |
| **external_port_range** | string |
| **logical_ip** | string |
| **logical_port** | optional string |
| **allowed_ext_ips** | optional **Address_Set** |
| **exempted_ext_ips** | optional **Address_Set** |
| **gateway_port** | optional weak reference to **Logical_Router_Port** |
| **match** | string |
| **priority** | integer, in range 0 to 32,767 |
| **options : stateless** | optional string |
| **options : stateless_icmp_helper** | optional string, either **true** or **false** |
| **options : add_route** | optional string |
| **options : dynamic-routing-advertise** | optional string |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**type**: string, one of **dnat**, **dnat_and_snat**, or **snat**

Type of the NAT rule.

- When **type** is **dnat**, the externally visible IP address **external_ip** is DNATted to the IP address **logical_ip** in the logical space.
- When **type** is **snat**, IP packets with their source IP address that either matches the IP address in **logical_ip** or is in the network provided by **logical_ip** is SNATed into the IP address in **external_ip**.
- When **type** is **dnat_and_snat**, the externally visible IP address **external_ip** is DNATted to the IP address **logical_ip** in the logical space. In addition, IP packets with the source IP address that matches **logical_ip** is SNATed into the IP address in **external_ip**.

**external_ip**: string

An IPv4 address.

**external_mac**: optional string

A MAC address.

This is only used on the gateway port on distributed routers. This must be specified in order for the NAT rule to be processed in a distributed manner on all chassis. If this is not specified for a NAT rule on a distributed router, then this NAT rule will be processed in a centralized manner on the gateway port instance on the gateway chassis.

This MAC address must be unique on the logical switch that the gateway port is attached to. If the MAC address used on the **logical_port** is globally unique, then that MAC address can be specified as this **external_mac**.

**external_port_range**: string

L4 source port range

Range of ports, from which a port number will be picked that will replace the source port of to be NATed packet. This is basically PAT (port address translation).

Value of the column is in the format, port_lo-port_hi. For example: external_port_range : "1–30000"

Valid range of ports is 1−65535.

**logical_ip**: string
An IPv4 network (e.g 192.168.1.0/24) or an IPv4 address.

**logical_port**: optional string
The name of the logical port where the **logical_ip** resides.

This is only used on distributed routers. This must be specified in order for the NAT rule to be processed in a distributed manner on all chassis. If this is not specified for a NAT rule on a distributed router, then this NAT rule will be processed in a centralized manner on the gateway port instance on the gateway chassis.

**allowed_ext_ips**: optional **Address_Set**
It represents Address Set of external ips that NAT rule is applicable to. For SNAT type NAT rules, this refers to destination addresses. For DNAT type NAT rules, this refers to source addresses.

This configuration overrides the default NAT behavior of applying a rule solely based on internal IP. Without this configuration, NAT happens without considering the external IP (i.e dest/source for snat/dnat type rule). With this configuration NAT rule is applied ONLY if external ip is in the input Address Set.

**exempted_ext_ips**: optional **Address_Set**
It represents Address Set of external ips that NAT rule is NOT applicable to. For SNAT type NAT rules, this refers to destination addresses. For DNAT type NAT rules, this refers to source addresses.

This configuration overrides the default NAT behavior of applying a rule solely based on internal IP. Without this configuration, NAT happens without considering the external IP (i.e dest/source for snat/dnat type rule). With this configuration NAT rule is NOT applied if external ip is in the input Address Set.

If there are NAT rules in a logical router with overlapping IP prefixes (including /32), then usage of *exempted_ext_ips* should be avoided in following scenario. a. SNAT rule (let us say RULE1) with logical_ip PREFIX/MASK (let us say 50.0.0.0/24). b. SNAT rule (let us say RULE2) with logical_ip PREFIX/MASK+1 (let us say 50.0.0.0/25). c. Now, if exempted_ext_ips is associated with RULE2, then a logical ip which matches both 50.0.0.0/24 and 50.0.0.0/25 may get the RULE2 applied to it instead of RULE1.

*allowed_ext_ips* and *exempted_ext_ips* are mutually exclusive to each other. If both Address Sets are set for a rule, then the NAT rule is not considered.

**gateway_port**: optional weak reference to **Logical_Router_Port**
A distributed gateway port in the **Logical_Router_Port** table where the NAT rule needs to be applied.

When multiple distributed gateway ports are configured on a **Logical_Router**, applying a NAT rule at each of the distributed gateway ports might not be desired. Consider the case where a logical router has 2 distributed gateway port, one with **networks 50.0.0.10/24** and the other with **networks 60.0.0.10/24**. If the logical router has a NAT rule of **type snat**, **logical_ip 10.1.1.0/24** and **external_ip 50.1.1.20/24**, the rule needs to be selectively applied on matching packets entering/leaving through the distributed gateway port with **networks 50.0.0.10/24**.

When a logical router has multiple distributed gateway ports and this column is not set for a NAT rule, then the rule will be applied at the distributed gateway port which is in the same network as the **external_ip** of the NAT rule, if such a router port exists. If logical router has a single distributed gateway port and this column is not set for a NAT rule, the rule will be applied at the distributed gateway port even if the router port is not in the same network as the **external_ip** of the NAT rule.

**match**: string
The packets that the NAT rules should match, in addition to the match that is created based on the NAT type, in the same expression language used for the **match** column in the OVN Southbound

database's **Logical_Flow** table. This allows for more fine-grained control over the NAT rule.

**priority**: integer, in range 0 to 32,767
The NAT rule's priority. Rules with numerically higher priority take precedence over those with lower. The priority is taken into account only if the **match** is defined.

**options : stateless**: optional string
Indicates if a dnat_and_snat rule should lead to connection tracking state or not.

**options : stateless_icmp_helper**: optional string, either **true** or **false**
Applies only to stateless **dnat_and_snat** rules (that is, NATs with **options:stateless** set to **true**) on IPv4 addresses. Defaults to **true**.

A stateless DNAT rule rewrites only the outer IPv4 destination of inbound packets. For an inbound ICMPv4 error (for example a **Fragmentation Needed** message generated for Path MTU discovery, RFC 1191, or a **Time Exceeded** message generated for a traceroute probe), the original packet embedded in the ICMP payload still carries the external, post-NAT IP as its source. When the error reaches the downstream logical switch, conntrack cannot correlate the embedded tuple with the tracked outgoing flow, the packet is marked **ct.inv** and dropped by the ACL stage, and PMTU discovery and traceroute break.

When this option is **true**, **ovn−northd** emits an additional, higher-priority logical flow in the router ingress DNAT stage that matches the ICMPv4 errors which quote the original datagram - Destination Unreachable (type 3, which includes **Fragmentation Needed**), Time Exceeded (type 11) and Parameter Problem (type 12) - destined to the external IP. It rewrites the outer IPv4 destination to the logical IP and, using the **icmp4.inner_ip4.src** action, un-NATs the embedded inner IPv4 source from the external IP back to the logical IP, so that conntrack can correlate the error. This makes PMTU discovery and traceroute work end-to-end. Set it to **false** to suppress this flow for an individual NAT entry.

**options : add_route**: optional string
If set to **true**, then neighbor routers will have logical flows added that will allow for routing to the NAT address. It also will have ARP resolution logical flows added. By setting this option, it means there is no reason to create a **Logical_Router_Static_Route** from neighbor routers to this NAT address. It also means that no ARP request is required for neighbor routers to learn the IP-MAC mapping for this NAT address. This option only applies to NATs of type **dnat** and **dnat_and_snat**. For more information about what flows are added for IP routes, please see the **ovn−northd** manpage section on IP Routing.

**options : dynamic-routing-advertise**: optional string
If the CMS sets **options:dynamic-routing-redistribute** to **nat**, this parameter is used by the CMS to enable/disable the advertisement of the current NAT external IP via the **Advertised_Route** table. Default: **true**.

For a distributed **dnat_and_snat** entry (a floating IP, with both **external_mac** and **logical_port** set) this also controls the EVPN Type−2 advertisement: when the provider **Logical_Switch** that carries the NAT's distributed gateway port has **nat** set in its **other_config:dynamic-routing-redistribute**, the external IP/MAC are published in the **Advertised_MAC_Binding** table. Setting this option to **false** opts the individual floating IP out of that advertisement as well.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

## DHCP_Options TABLE

OVN implements native DHCPv4 support which caters to the common use case of providing an IPv4 address to a booting instance by providing stateless replies to DHCPv4 requests based on statically configured address mappings. To do this it allows a short list of DHCPv4 options to be configured and applied at each compute host running **ovn−controller**.

OVN also implements native DHCPv6 support which provides stateless replies to DHCPv6 requests.

**Summary:**

| | |
|---|---|
| **cidr** | string |
| *DHCPv4 options:* | |
| *Mandatory DHCPv4 options:* | |
| **options : server_id** | optional string |
| **options : server_mac** | optional string |
| **options : lease_time** | optional string, containing an integer, in range 0 to 4,294,967,295 |
| *IPv4 DHCP Options:* | |
| **options : router** | optional string |
| **options : netmask** | optional string |
| **options : dns_server** | optional string |
| **options : log_server** | optional string |
| **options : lpr_server** | optional string |
| **options : swap_server** | optional string |
| **options : policy_filter** | optional string |
| **options : router_solicitation** | optional string |
| **options : nis_server** | optional string |
| **options : ntp_server** | optional string |
| **options : netbios_name_server** | optional string |
| **options : classless_static_route** | optional string |
| **options : ms_classless_static_route** | optional string |
| **options : next_server** | optional string |
| *Boolean DHCP Options:* | |
| **options : ip_forward_enable** | optional string, either **0** or **1** |
| **options : router_discovery** | optional string, either **0** or **1** |
| **options : ethernet_encap** | optional string, either **0** or **1** |
| *Integer DHCP Options:* | |
| **options : default_ttl** | optional string, containing an integer, in range 0 to 255 |
| **options : tcp_ttl** | optional string, containing an integer, in range 0 to 255 |
| **options : mtu** | optional string, containing an integer, in range 68 to 65,535 |
| **options : T1** | optional string, containing an integer, in range 68 to 4,294,967,295 |
| **options : T2** | optional string, containing an integer, in range 68 to 4,294,967,295 |
| **options : arp_cache_timeout** | optional string, containing an integer, in range 0 to 255 |
| **options : tcp_keepalive_interval** | optional string, containing an integer, in range 0 to 255 |
| **options : netbios_node_type** | optional string, containing an integer, in range 0 to 255 |
| *String DHCP Options:* | |
| **options : wpad** | optional string |
| **options : bootfile_name** | optional string |
| **options : path_prefix** | optional string |
| **options : tftp_server_address** | optional string |

| | |
|---|---|
| **options : hostname** | optional string |
| **options : domain_name** | optional string |
| **options : bootfile_name_alt** | optional string |
| **options : broadcast_address** | optional string |
| *DHCP Options of type host_id:* | |
| **options : tftp_server** | optional string |
| *DHCP Options of type domains:* | |
| **options : domain_search_list** | optional string |
| *DHCPv6 options:* | |
| *Mandatory DHCPv6 options:* | |
| **options : server_id** | optional string |
| *IPv6 DHCPv6 options:* | |
| **options : dns_server** | optional string |
| *String DHCPv6 options:* | |
| **options : domain_search** | optional string |
| **options : dhcpv6_stateless** | optional string |
| **options : fqdn** | optional string |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**cidr**: string

The DHCPv4/DHCPv6 options will be included if the logical port has its IP address in this **cidr**.

*DHCPv4 options:*

The CMS should define the set of DHCPv4 options as key/value pairs in the **options** column of this table. For **ovn−controller** to include these DHCPv4 options, the **dhcpv4_options** of **Logical_Switch_Port** should refer to an entry in this table.

*Mandatory DHCPv4 options:*

The following options must be defined.

**options : server_id**: optional string

The IP address for the DHCP server to use. This should be in the subnet of the offered IP. This is also included in the DHCP offer as option 54, ‘‘server identifier.’’

**options : server_mac**: optional string

The Ethernet address for the DHCP server to use.

**options : lease_time**: optional string, containing an integer, in range 0 to 4,294,967,295

The offered lease time in seconds,

The DHCPv4 option code for this option is 51.

*IPv4 DHCP Options:*

Below are the supported DHCPv4 options whose values are an IPv4 address, e.g. **192.168.1.1**. Some options accept multiple IPv4 addresses enclosed within curly braces, e.g. **{192.168.1.2, 192.168.1.3}**. Please refer to RFC 2132 for more details on DHCPv4 options and their codes.

**options : router**: optional string

The IP address of a gateway for the client to use. This should be in the subnet of the offered IP. The DHCPv4 option code for this option is 3.

**options : netmask**: optional string

The DHCPv4 option code for this option is 1.

**options : dns_server**: optional string
The DHCPv4 option code for this option is 6.

**options : log_server**: optional string
The DHCPv4 option code for this option is 7.

**options : lpr_server**: optional string
The DHCPv4 option code for this option is 9.

**options : swap_server**: optional string
The DHCPv4 option code for this option is 16.

**options : policy_filter**: optional string
The DHCPv4 option code for this option is 21.

**options : router_solicitation**: optional string
The DHCPv4 option code for this option is 32.

**options : nis_server**: optional string
The DHCPv4 option code for this option is 41.

**options : ntp_server**: optional string
The DHCPv4 option code for this option is 42.

**options : netbios_name_server**: optional string
The DHCPv4 option code for this option is 44.

**options : classless_static_route**: optional string
The DHCPv4 option code for this option is 121.

This option can contain one or more static routes, each of which consists of a destination descriptor and the IP address of the router that should be used to reach that destination. Please see RFC 3442 for more details.

Example: **{30.0.0.0/24,10.0.0.10, 0.0.0.0/0,10.0.0.1}**

**options : ms_classless_static_route**: optional string
The DHCPv4 option code for this option is 249. This option is similar to **classless_static_route** supported by Microsoft Windows DHCPv4 clients.

**options : next_server**: optional string
The DHCPv4 option code for setting the "Next server IP address" field in the DHCP header.

*Boolean DHCP Options:*

These options accept a Boolean value, expressed as **0** for false or **1** for true.

**options : ip_forward_enable**: optional string, either **0** or **1**
The DHCPv4 option code for this option is 19.

**options : router_discovery**: optional string, either **0** or **1**
The DHCPv4 option code for this option is 31.

**options : ethernet_encap**: optional string, either **0** or **1**
The DHCPv4 option code for this option is 36.

*Integer DHCP Options:*

These options accept a nonnegative integer value.

**options : default_ttl**: optional string, containing an integer, in range 0 to 255
The DHCPv4 option code for this option is 23.

**options : tcp_ttl**: optional string, containing an integer, in range 0 to 255
The DHCPv4 option code for this option is 37.

**options : mtu**: optional string, containing an integer, in range 68 to 65,535
The DHCPv4 option code for this option is 26.

**options : T1**: optional string, containing an integer, in range 68 to 4,294,967,295
This specifies the time interval from address assignment until the client begins trying to renew its address. The DHCPv4 option code for this option is 58.

**options : T2**: optional string, containing an integer, in range 68 to 4,294,967,295
This specifies the time interval from address assignment until the client begins trying to rebind its address. The DHCPv4 option code for this option is 59.

**options : arp_cache_timeout**: optional string, containing an integer, in range 0 to 255
The DHCPv4 option code for this option is 35. This option specifies the timeout in seconds for ARP cache entries.

**options : tcp_keepalive_interval**: optional string, containing an integer, in range 0 to 255
The DHCPv4 option code for this option is 38. This option specifies the interval that the client TCP should wait before sending a keepalive message on a TCP connection.

**options : netbios_node_type**: optional string, containing an integer, in range 0 to 255
The DHCPv4 option code for this option is 46.

*String DHCP Options:*

These options accept a string value.

**options : wpad**: optional string
The DHCPv4 option code for this option is 252. This option is used as part of web proxy auto discovery to provide a URL for a web proxy.

**options : bootfile_name**: optional string
The DHCPv4 option code for this option is 67. This option is used to identify a bootfile.

**options : path_prefix**: optional string
The DHCPv4 option code for this option is 210. In PXELINUX' case this option is used to set a common path prefix, instead of deriving it from the bootfile name.

**options : tftp_server_address**: optional string
The DHCPv4 option code for this option is 150. The option contains one or more IPv4 addresses that the client MAY use. This option is Cisco proprietary, the IEEE standard that matches with this requirement is option 66 (tftp_server).

**options : hostname**: optional string
The DHCPv4 option code for this option is 12. If set, indicates the DHCPv4 option "Hostname". Alternatively, this option can be configured in **options:hostname** column in table **Logical_Switch_Port**. If Hostname option value is set in both conflicting **Logical_Switch_Port** and **DHCP_Options** tables, **Logical_Switch_Port** takes precedence.

**options : domain_name**: optional string
The DHCPv4 option code for this option is 15. This option specifies the domain name that client should use when resolving hostnames via the Domain Name System.

**options : bootfile_name_alt**: optional string
"bootfile_name_alt" option is used to support iPXE. When both "bootfile_name" and "bootfile_name_alt" are provided by the CMS, "bootfile_name" will be used for option 67 if the dhcp request contains etherboot option (175), otherwise "bootfile_name_alt" will be used.

**options : broadcast_address**: optional string
The DHCPv4 option code for this option is 28. This option specifies the IP address used as a broadcast address.

*DHCP Options of type host_id:*

These options accept either an IPv4 address or a string value.

**options : tftp_server**: optional string
The DHCPv4 option code for this option is 66.

*DHCP Options of type domains:*

These options accept string value which is a comma separated list of domain names. The domain names are encoded based on RFC 1035.

**options : domain_search_list**: optional string
The DHCPv4 option code for this option is 119.

*DHCPv6 options:*

OVN also implements native DHCPv6 support. The CMS should define the set of DHCPv6 options as key/value pairs. The define DHCPv6 options will be included in the DHCPv6 response to the DHCPv6 Solicit/Request/Confirm packet from the logical ports having the IPv6 addresses in the **cidr**.

*Mandatory DHCPv6 options:*

The following options must be defined.

**options : server_id**: optional string
The Ethernet address for the DHCP server to use. This is also included in the DHCPv6 reply as option 2, ‘‘Server Identifier’’ to carry a DUID identifying a server between a client and a server. **ovn−controller** defines DUID based on Link-layer Address [DUID-LL].

*IPv6 DHCPv6 options:*

Below are the supported DHCPv6 options whose values are an IPv6 address, e.g. **aef0::4**. Some options accept multiple IPv6 addresses enclosed within curly braces, e.g. **{aef0::4, aef0::5}**. Please refer to RFC 3315 for more details on DHCPv6 options and their codes.

**options : dns_server**: optional string
The DHCPv6 option code for this option is 23. This option specifies the DNS servers that the VM should use.

*String DHCPv6 options:*

These options accept string values.

**options : domain_search**: optional string
The DHCPv6 option code for this option is 24. This option specifies the domain search list the client should use to resolve hostnames with DNS.

Example: **"ovn.org"**.

**options : dhcpv6_stateless**: optional string
This option specifies the OVN native DHCPv6 will work in stateless mode, which means OVN native DHCPv6 will not offer IPv6 addresses for VM/VIF ports, but only reply other configurations, such as DNS and domain search list. When setting this option with string value "true", VM/VIF will configure IPv6 addresses by stateless way. Default value for this option is false.

**options : fqdn**: optional string
The DHCPv6 option code for this option is 39. If set, indicates the DHCPv6 option "FQDN".

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

### DHCP_Relay TABLE

OVN implements native DHCPv4 relay support which caters to the common use case of relaying the DHCP requests to external DHCP server.

**Summary:**

| | |
|---|---|
| **name** | string |
| **servers** | optional string |
| **options** | map of string-string pairs |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string
: A name for the DHCP Relay.

**servers**: optional string
: The DHCPv4 server IP address.

**options**: map of string-string pairs
: Future purpose.

*Common Columns:*

**external_ids**: map of string-string pairs
: See **External IDs** at the beginning of this document.

## Connection TABLE

Configuration for a database connection to an Open vSwitch database (OVSDB) client.

This table primarily configures the Open vSwitch database server (**ovsdb−server**).

The Open vSwitch database server can initiate and maintain active connections to remote clients. It can also listen for database connections.

**Summary:**

| | |
|---|---|
| *Core Features:* | |
| **target** | string (must be unique within table) |
| *Client Failure Detection and Handling:* | |
| **max_backoff** | optional integer, at least 1,000 |
| **inactivity_probe** | optional integer |
| *Status:* | |
| **is_connected** | boolean |
| **status : last_error** | optional string |
| **status : state** | optional string, one of **ACTIVE**, **BACKOFF**, **CONNECTING**, **IDLE**, or **VOID** |
| **status : sec_since_connect** | optional string, containing an integer, at least 0 |
| **status : sec_since_disconnect** | optional string, containing an integer, at least 0 |
| **status : locks_held** | optional string |
| **status : locks_waiting** | optional string |
| **status : locks_lost** | optional string |
| **status : n_connections** | optional string, containing an integer, at least 2 |
| **status : bound_port** | optional string, containing an integer |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |
| **other_config** | map of string-string pairs |

**Details:**

*Core Features:*

**target**: string (must be unique within table)

Connection methods for clients.

The following connection methods are currently supported:

**ssl:***host*[**:***port*]

The specified SSL/TLS *port* on the host at the given *host*, which can either be a DNS name (if built with unbound library) or an IP address. A valid SSL/TLS configuration must be provided when this form is used, this configuration can be specified via command-line options or the **SSL** table.

If *port* is not specified, it defaults to 6640.

SSL/TLS support is an optional feature that is not always built as part of OVN or Open vSwitch.

**tcp:***host*[**:***port*]

The specified TCP *port* on the host at the given *host*, which can either be a DNS name (if built with unbound library) or an IP address. If *host* is an IPv6 address, wrap it in square brackets, e.g. **tcp:[::1]:6640**.

If *port* is not specified, it defaults to 6640.

**pssl:**[*port*][**:***host*]

Listens for SSL/TLS connections on the specified TCP *port*. Specify 0 for *port* to have the kernel automatically choose an available port. If *host*, which can either be a DNS name (if built with unbound library) or an IP address, is specified, then connections are restricted to

the resolved or specified local IPaddress (either IPv4 or IPv6 address). If *host* is an IPv6 address, wrap in square brackets, e.g. **pssl:6640:[::1]**. If *host* is not specified then it listens only on IPv4 (but not IPv6) addresses. A valid SSL/TLS configuration must be provided when this form is used, this can be specified either via command-line options or the **SSL** table.

If *port* is not specified, it defaults to 6640.

SSL/TLS support is an optional feature that is not always built as part of OVN or Open vSwitch.

**ptcp:**[*port*][**:***host*]
Listens for connections on the specified TCP *port*. Specify 0 for *port* to have the kernel automatically choose an available port. If *host*, which can either be a DNS name (if built with unbound library) or an IP address, is specified, then connections are restricted to the resolved or specified local IP address (either IPv4 or IPv6 address). If *host* is an IPv6 address, wrap it in square brackets, e.g. **ptcp:6640:[::1]**. If *host* is not specified then it listens only on IPv4 addresses.

If *port* is not specified, it defaults to 6640.

When multiple clients are configured, the **target** values must be unique. Duplicate **target** values yield unspecified results.

*Client Failure Detection and Handling:*

**max_backoff**: optional integer, at least 1,000
Maximum number of milliseconds to wait between connection attempts. Default is implementation-specific.

**inactivity_probe**: optional integer
Maximum number of milliseconds of idle time on connection to the client before sending an inactivity probe message. If Open vSwitch does not communicate with the client for the specified number of seconds, it will send a probe. If a response is not received for the same additional amount of time, Open vSwitch assumes the connection has been broken and attempts to reconnect. Default is implementation-specific. A value of 0 disables inactivity probes.

*Status:*

Key-value pair of **is_connected** is always updated. Other key-value pairs in the status columns may be updated depends on the **target** type.

When **target** specifies a connection method that listens for inbound connections (e.g. **ptcp:** or **punix:**), both **n_connections** and **is_connected** may also be updated while the remaining key-value pairs are omitted.

On the other hand, when **target** specifies an outbound connection, all key-value pairs may be updated, except the above-mentioned two key-value pairs associated with inbound connection targets. They are omitted.

**is_connected**: boolean
**true** if currently connected to this client, **false** otherwise.

**status : last_error**: optional string
A human-readable description of the last error on the connection to the manager; i.e. **strerror(errno)**. This key will exist only if an error has occurred.

**status : state**: optional string, one of **ACTIVE**, **BACKOFF**, **CONNECTING**, **IDLE**, or **VOID**
The state of the connection to the manager:

**VOID** Connection is disabled.

**BACKOFF**
Attempting to reconnect at an increasing period.

**CONNECTING**
Attempting to connect.

**ACTIVE**
Connected, remote host responsive.

**IDLE** Connection is idle. Waiting for response to keep-alive.

These values may change in the future. They are provided only for human consumption.

**status : sec_since_connect**: optional string, containing an integer, at least 0
The amount of time since this client last successfully connected to the database (in seconds). Value is empty if client has never successfully been connected.

**status : sec_since_disconnect**: optional string, containing an integer, at least 0
The amount of time since this client last disconnected from the database (in seconds). Value is empty if client has never disconnected.

**status : locks_held**: optional string
Space-separated list of the names of OVSDB locks that the connection holds. Omitted if the connection does not hold any locks.

**status : locks_waiting**: optional string
Space-separated list of the names of OVSDB locks that the connection is currently waiting to acquire. Omitted if the connection is not waiting for any locks.

**status : locks_lost**: optional string
Space-separated list of the names of OVSDB locks that the connection has had stolen by another OVSDB client. Omitted if no locks have been stolen from this connection.

**status : n_connections**: optional string, containing an integer, at least 2
When **target** specifies a connection method that listens for inbound connections (e.g. **ptcp:** or **pssl:**) and more than one connection is actually active, the value is the number of active connections. Otherwise, this key-value pair is omitted.

**status : bound_port**: optional string, containing an integer
When **target** is **ptcp:** or **pssl:**, this is the TCP port on which the OVSDB server is listening. (This is particularly useful when **target** specifies a port of 0, allowing the kernel to choose any available port.)

*Common Columns:*

The overall purpose of these columns is described under **Common Columns** at the beginning of this document.

**external_ids**: map of string-string pairs

**other_config**: map of string-string pairs

## DNS TABLE

Each row in this table stores the DNS records. The **Logical_Switch** table's **dns_records** references these records.

**Summary:**

| | |
|---|---|
| **records** | map of string-string pairs |
| **options : ovn-owned** | optional string |
| **external_ids** | map of string-string pairs |

**Details:**

**records**: map of string-string pairs

Key-value pair of DNS records with **DNS query name** as the key and value as a string of IP address(es) separated by comma or space. For PTR requests, the key-value pair can be **Reverse IPv4 address.in−addr.arpa** and the value **DNS domain name**. For IPv6 addresses, the key has to be **Reverse IPv6 address.ip6.arpa**.

**Example:** "vm1.ovn.org" = "10.0.0.4 aef0::4"

**Example:** "4.0.0.10.in-addr.arpa" = "vm1.ovn.org"

**options : ovn-owned**: optional string

If set to true, then the OVN will be the main responsible for **DNS Records** within this row.

A **DNS** row with this option set to **true** can be created for domains that the user needs to configure locally and don't care about IPv6 only interested in IPv4 or vice versa. This will let ovn send IPv4 DNS reply and reject/ignore IPv6 queries to save the waiting for a timeout on those uninteresting queries.

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

## SSL TABLE

SSL/TLS configuration for ovn-nb database access.

**Summary:**

| | |
|---|---|
| **private_key** | string |
| **certificate** | string |
| **ca_cert** | string |
| **bootstrap_ca_cert** | boolean |
| **ssl_protocols** | string |
| **ssl_ciphers** | string |
| **ssl_ciphersuites** | string |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**private_key**: string

Name of a PEM file containing the private key used as the switch's identity for SSL/TLS connections to the controller.

**certificate**: string

Name of a PEM file containing a certificate, signed by the certificate authority (CA) used by the controller and manager, that certifies the switch's private key, identifying a trustworthy switch.

**ca_cert**: string

Name of a PEM file containing the CA certificate used to verify that the switch is connected to a trustworthy controller.

**bootstrap_ca_cert**: boolean

If set to **true**, then Open vSwitch will attempt to obtain the CA certificate from the controller on its first SSL/TLS connection and save it to the named PEM file. If it is successful, it will immediately drop the connection and reconnect, and from then on all SSL/TLS connections must be authenticated by a certificate signed by the CA certificate thus obtained. **This option exposes the SSL/TLS connection to a man−in−the−middle attack obtaining the initial CA certificate.** It may still be useful for bootstrapping.

**ssl_protocols**: string

Range or a comma- or space-delimited list of the SSL/TLS protocols to enable for SSL/TLS connections.

Supported protocols include **TLSv1.2** and **TLSv1.3**. Ranges can be provided in a form of two protocol names separated with a dash (**TLSv1.2−TLSv1.3**), or as a single protocol name with a plus sign (**TLSv1.2+**). The value can be a list of protocols or exactly one range. The range is a preferred way of specifying protocols and the configuration always behaves as if the range between the minimum and the maximum specified version is provided, i.e., if the value is set to **TLSv1.X,TLSv1.(X+2)**, the **TLSv1.(X+1)** will also be enabled as if it was a range. Regardless of order, the highest protocol supported by both sides will be chosen when making the connection.

The default when this option is omitted is **TLSv1.2+**.

**ssl_ciphers**: string

List of ciphers (in OpenSSL cipher string format) to be supported for SSL/TLS connections with TLSv1.2. The default when this option is omitted is **DEFAULT:@SECLEVEL=2**.

**ssl_ciphersuites**: string

List of ciphersuites (in OpenSSL ciphersuites string format) to be supported for SSL/TLS connections with TLSv1.3 and later. Default value from OpenSSL will be used when this option is omitted.

*Common Columns:*

The overall purpose of these columns is described under **Common Columns** at the beginning of this

document.

**external_ids**: map of string-string pairs

### Gateway_Chassis TABLE

Association of a chassis to a logical router port. The traffic going out through an specific router port will be redirected to a chassis, or a set of them in high availability configurations.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **chassis_name** | string |
| **priority** | integer, in range 0 to 32,767 |
| **options** | map of string-string pairs |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)

Name of the **Gateway_Chassis**.

A suggested, but not required naming convention is **${port_name}_${chassis_name}**.

**chassis_name**: string

Name of the chassis that we want to redirect traffic through for the associated logical router port. The value must match the **name** column of the **Chassis** table in the **OVN_Southbound** database.

**priority**: integer, in range 0 to 32,767

This is the priority of a chassis among all **Gateway_Chassis** belonging to the same logical router port.

**options**: map of string-string pairs

Reserved for future use.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### HA_Chassis_Group TABLE

Table representing a group of chassis which can provide high availability services. Each chassis in the group is represented by the table **HA_Chassis**. The HA chassis with highest priority will be the active chassis of this group. If the active chassis failover is detected, the HA chassis with the next higher priority takes over the responsibility of providing the HA. If a distributed gateway router port references a row in this table, then the active HA chassis in this group provides the gateway functionality.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **ha_chassis** | set of **HA_Chassis**es |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)

Name of the **HA_Chassis_Group**. Name should be unique.

**ha_chassis**: set of **HA_Chassis**es

A list of HA chassis which belongs to this group.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### HA_Chassis TABLE

**Summary:**

| | |
|---|---|
| **chassis_name** | string |
| **priority** | integer, in range 0 to 32,767 |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**chassis_name**: string

Name of the chassis which is part of the HA chassis group. The value must match the **name** column of the **Chassis** table in the **OVN_Southbound** database.

**priority**: integer, in range 0 to 32,767

Priority of the chassis. Chassis with highest priority will be the active chassis.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

## BFD TABLE

Contains BFD parameter for ovn-controller BFD configuration. OVN BFD implementation is used to provide detection of failures in the path between adjacent forwarding engines, including the OVN interfaces. OVN BFD provides link status info to OVN northd in order to update logical flows according to the status of BFD endpoints. In the current implementation OVN BFD is used to check next-hop status for ECMP routes. Please note BFD table refers to OVN BFD implementation and not to OVS legacy one.

**Summary:**

*Configuration:*

| | |
|---|---|
| **logical_port** | string |
| **dst_ip** | string |
| **min_tx** | optional integer, at least 1 |
| **min_rx** | optional integer |
| **detect_mult** | optional integer, at least 1 |
| **options** | map of string-string pairs |
| **external_ids** | map of string-string pairs |

*Status Reporting:*

| | |
|---|---|
| **status** | optional string, one of **admin_down**, **down**, **init**, or **up** |

**Details:**

*Configuration:*

**ovn−northd** reads configuration from these columns.

**logical_port**: string
: OVN logical port when BFD engine is running.

**dst_ip**: string
: BFD peer IP address.

**min_tx**: optional integer, at least 1
: This is the minimum interval, in milliseconds, that the local system would like to use when transmitting BFD Control packets, less any jitter applied. The value zero is reserved. Default value is 1000 ms.

**min_rx**: optional integer
: This is the minimum interval, in milliseconds, between received BFD Control packets that this system is capable of supporting, less any jitter applied by the sender. If this value is zero, the transmitting system does not want the remote system to send any periodic BFD Control packets.

**detect_mult**: optional integer, at least 1
: Detection time multiplier. The negotiated transmit interval, multiplied by this value, provides the Detection Time for the receiving system in Asynchronous mode. Default value is 5.

**options**: map of string-string pairs
: Reserved for future use.

**external_ids**: map of string-string pairs
: See **External IDs** at the beginning of this document.

*Status Reporting:*

**ovn−northd** writes BFD status into these columns.

**status**: optional string, one of **admin_down**, **down**, **init**, or **up**
: BFD port logical states. Possible values are:

  - **admin_down**
  - **down**

- **init**
- **up**

### Static_MAC_Binding TABLE

Each record represents a Static_MAC_Binding entry for a logical router.

**Summary:**

*Configuration:*

| | |
|---|---|
| **logical_port** | string |
| **ip** | string |
| **mac** | string |
| **override_dynamic_mac** | boolean |

**Details:**

*Configuration:*

**ovn−northd** reads configuration from these columns and propagates the value to SBDB.

**logical_port**: string
: The logical router port for the binding.

**ip**: string
: The bound IP address.

**mac**: string
: The Ethernet address to which the IP is bound.

**override_dynamic_mac**: boolean
: Override dynamically learnt MACs.

### Chassis_Template_Var TABLE

One record per chassis, each containing a map, **variables**, between template variable names and their value for that specific chassis. A template variable has a name and potentially different values on different hypervisors in the OVN cluster. For example, two rows, **R1 = (.chassis=C1, variables={(N: V1)}** and **R2 = (.chassis=C2, variables={(N: V2)}** will make **ovn−controller** running on chassis **C1** and **C2** interpret the token **N** either as **V1** (on **C1**) or as **V2** (on **C2**). Users can refer to template variables from within other logical components, e.g., within ACL, QoS or Logical_Router_Policy matches or from Load_Balancer VIP and backend definitions.

If a template variable is referenced on a chassis for which that variable is not defined then **ovn−controller** running on that chassis will just interpret it as a raw string literal.

**Summary:**

| | |
|---|---|
| **chassis** | string (must be unique within table) |
| **variables** | map of string-string pairs |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**chassis**: string (must be unique within table)

The chassis this set of variable values applies to.

**variables**: map of string-string pairs

The set of variable values for a given chassis.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

## Sampling_App TABLE

**Summary:**

| | |
|---|---|
| **type** | string, one of **acl−est**, **acl−new**, or **drop** (must be unique within table) |
| **id** | integer, in range 1 to 255 |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**type**: string, one of **acl−est**, **acl−new**, or **drop** (must be unique within table)

The type of the application to be configured for sampling. Currently supported options are: "drop", "acl-new", "acl-est".

**id**: integer, in range 1 to 255

The identifier to be encoded in the samples generated for this type of application. This identifier is used as part of the sample's observation domain ID.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

## Network_Function_Group TABLE

Each row contains a list of **Network_Function**. Traffic redirection/mirroring is achieved by referencing a **Network_Function_Group** from an **ACL**. For **inline** mode, health monitoring of each **Network_Function** is performed based on parameters defined in **Network_Function_Health_Check**. Health monitoring is not applicable for **vtap** mode; a vtap network function is always considered active as long as its port is bound. Traffic matching the ACL is redirected (inline) or mirrored (vtap) to one of the active **Network_Functions**. If all are detected as down, the fallback method is applied.

**Known limitation (inline mode)**: in topologies where a post-NF flood copy may reach the same logical inport from a different node than the packet's original ingress node, the lookup runs on a different node than where the inport was learned, so loopback copies are not identified. For example: OVN-IC L2 stretch with the NF and IC gateway on different chassis. With **unknown**-addressed ports, the post-NF packet can be flooded across AZs and come back to the source AZ via another node, so the lookup does not hit the original learn entry.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **id** | integer, in range 1 to 255 (must be unique within table) |
| **fallback** | optional string, either **fail−close** or **fail−open** |
| **network_function** | set of **Network_Function**s |
| **network_function_active** | optional **Network_Function** |
| **mode** | string, either **inline** or **vtap** |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)

Name of the **Network_Function_Group**. Name should be unique.

**id**: integer, in range 1 to 255 (must be unique within table)

A unique integer between 1 and 255 must be assigned to each **Network_Function_Group**.

**fallback**: optional string, either **fail−close** or **fail−open**

Fallback setting when no active network functions are available.

Supports following fallback mechanisms. If not specified: for **inline** mode, fail-close is applied when no active Network Functions are available; for **vtap** mode, fallback is always fail-open.

**fail−open**

Traffic bypasses Network Function and gets allowed when there is no active Network Function available.

**fail−close**

Traffic gets dropped when there is no active Network Function available.

**network_function**: set of **Network_Function**s

A list of network functions which belong to this group.

**network_function_active**: optional **Network_Function**

Current active Network_Function. This column is populated by northd based on health monitoring status for **inline** mode. For **vtap** mode, health monitoring is not applicable; northd sets the active network function based on port binding status.

**mode**: string, either **inline** or **vtap**

Network Function insertion mode.

Supports following modes.

**inline** In this mode, traffic redirection happens inline through active Network Function ports.

**vtap** In this mode, traffic is mirrored to active Network Function port.

*Common Columns:*

**external_ids**: map of string-string pairs
See **External IDs** at the beginning of this document.

### Network_Function TABLE

Each row represents one network function entity. This contains a single logical switch port when deployed in vtap mode and pair of logical switch ports when deployed in inline mode. In inline mode, traffic that matches the ACL is redirected to **inport** for from-lport ACLs and to **outport** for to-lport ACLs. Once the traffic is received on the other port, it continues through the standard OVN pipeline. Response traffic follows the reverse path: it is redirected to the **outport** for from-lport ACLs and to the **inport** for to-lport ACLs. Once the traffic is received on the other port, it is processed by the regular OVN pipeline. **NOTE**: 1. The Network Function MUST NOT modify the packet headers. 2. The Network Function is not supported when used in conjunction with Load Balancer. In vtap mode, traffic that matches the ACL is mirrored to **inport** in all cases.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| **id** | integer, in range 1 to 255 (must be unique within table) |
| **inport** | **Logical_Switch_Port** |
| **outport** | optional **Logical_Switch_Port** |
| **health_check** | optional **Network_Function_Health_Check** |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)

Name of the **Network_Function**. Name should be unique.

**id**: integer, in range 1 to 255 (must be unique within table)

A unique integer between 1 and 255 must be assigned to each **Network_Function**.

**inport**: **Logical_Switch_Port**

**Logical_Switch_Port** where request traffic for from-lport ACL and response traffic for to-lport ACL is redirected when network function is deployed in inline mode. In vtap mode, traffic matching both to-lport and from-lport ACLs is mirrored to the inport.

**outport**: optional **Logical_Switch_Port**

**Logical_Switch_Port** where request traffic for to-lport ACL and response traffic for from-lport ACL is redirected when network function is deployed in inline mode. This is not applicable when network function is deployed in vtap mode.

**health_check**: optional **Network_Function_Health_Check**

**Network_Function_Health_Check** associated with this network function. Health check is applicable only for **inline** mode. It is not applicable when the network function is deployed in **vtap** mode; this column should be left unset for vtap network functions.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.

### Network_Function_Health_Check TABLE

Each row represents one network function health check.

**Summary:**

| | |
|---|---|
| **name** | string (must be unique within table) |
| *Health check options:* | |
| **options : interval** | optional string, containing an integer |
| **options : timeout** | optional string, containing an integer |
| **options : success_count** | optional string, containing an integer |
| **options : failure_count** | optional string, containing an integer |
| *Common Columns:* | |
| **external_ids** | map of string-string pairs |

**Details:**

**name**: string (must be unique within table)

Name of the **Network_Function_Health_Check**. Name should be unique.

*Health check options:*

**options : interval**: optional string, containing an integer

The interval, in seconds, between health checks. Default: 5s.

**options : timeout**: optional string, containing an integer

The time, in seconds, after which a health check times out. Default: 3s.

**options : success_count**: optional string, containing an integer

The number of successful checks after which the Network_Function is considered online. Default: 1.

**options : failure_count**: optional string, containing an integer

The number of failure checks after which the Network_Function is considered offline. Default: 1.

*Common Columns:*

**external_ids**: map of string-string pairs

See **External IDs** at the beginning of this document.